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AI Driven Drug Repurpose

A thesis submitted by

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A dissertation submitted in partial fulfillment of the requirements for the degree of
Digilians 9 Months Diploma in Applied AI and Data Analytics

Under supervision of

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Military Technical College

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(August 2026)



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Abstract

Drug repurposing offers a strategically important alternative to de novo drug discovery by identifying new therapeutic indications for existing approved compounds, reducing development time, cost, and clinical attrition risk. However, systematic repurposing remains challenging because known drug–disease associations are extremely sparse (0.36% of the compound–disease search space in Hetionet v1.0), biomedical data are highly heterogeneous, and many predictive models lack interpretability. This project presents an AI-driven drug repurposing framework based on Hetionet, a heterogeneous biomedical knowledge graph integrating 29 databases, 11 node types, 24 relation types, 47,031 nodes, and over 2.25 million edges. To reduce single-model dependency, the framework combines two architecturally independent predictive components. The first, DREAMwalk with XGBoost, generates 128-dimensional node embeddings through semantic information-guided random walks of length 100, with 20 walks per node and teleportation probability ($p=0.5$). Transition probabilities are derived from vectorized Jaccard similarity matrices computed for each type of entity. Walk sequences are encoded using heterogeneous Skip-gram Word2Vec, while drug–disease pairs are represented by 256-dimensional concatenated embeddings and classified using XGBoost with 300 estimators, maximum depth 6, and learning rate 0.001. Evaluation follows a leak-free 10-fold stratified cross-validation protocol. The second component, TxGNN, is a heterogeneous graph neural network trained end-to-end using PyTorch Geometric. It employs 128-dimensional type-specific learnable node embeddings, a two-layer HeteroConv encoder with relation-specific GraphSAGE operators, and DistMult link scoring for CtD and CpD relations. Training uses BCEWithLogitsLoss, dynamic negative sampling, and auxiliary contrastive metric loss, with leak-free 5-fold stratified cross-validation. The two models are integrated through a weighted ensemble, assigning 55% to DREAMwalk/XGBoost and 45% to TxGNN. Their complementary learning mechanisms—walk-based co-occurrence and iterative graph message passing—provide diverse representations and reduce false-positive predictions. Interpretability is achieved through SHAP-based feature attribution over the XGBoost classifier, presented through interactive waterfall charts in a web-based decision-support interface supporting both Drug-to-Disease and Disease-to-Drug queries. Predictions are ranked by ensemble confidence and enriched with supporting gene sets, direct Hetionet evidence, and biological pathway information, while an XAI assistant converts SHAP outputs into human-readable explanations.

The DREAMwalk/XGBoost pipeline achieves 0.938 AUROC and 0.873 accuracy on the Hetionet benchmark, representing a 16.8% improvement over state-of-the-art baselines. Real-world case studies targeting Alzheimer’s Disease and Breast Carcinoma further demonstrate practical applicability. Overall, the framework demonstrates how heterogeneous knowledge graphs, semantic random walks, graph neural networks, ensemble learning, and explainable AI can enable accurate, systematic, and interpretable drug repurposing.

Acknowledgments

We would like to express our sincere gratitude to Allah for giving us the strength, patience, and guidance to complete this project successfully. We would like to extend our deepest appreciation to **Maj. Gen. Prof. Dr. Gouda Ismail** for his valuable supervision, continuous guidance, constructive feedback, and encouragement throughout the development of this project. His expertise and support greatly contributed to the progress and successful completion of our work. We would also like to express our sincere gratitude to **Eng. Omar Mamdouh** for his valuable technical guidance, continuous support, and helpful insights throughout the implementation and development of the project. We are deeply grateful to the **Military Technical College** and all our professors for providing us with the knowledge, skills, and academic environment that supported us throughout our studies. Finally, we would like to thank all our team members for their dedication, cooperation, and teamwork, which played an essential role in overcoming challenges and completing this project successfully. To everyone who supported us directly or indirectly, we sincerely express our gratitude.

AI-Driven Drug Repurpose

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Abbreviations

- **AI** – Artificial Intelligence
- **ATC** – Anatomical Therapeutic Chemical Classification
- **AUROC / AUPRC** – Area Under ROC / Precision-Recall Curve
- **BKG / biomedKG** – Biomedical Knowledge Graph
- **CtD / CpD** – Compound-treats-Disease / Compound-palliates-Disease
- **DREAMwalk** – Drug Repurposing via Multi-layer Random Walk

- **GBA** – Guilt-by-Association
 - **GNN / RGCN / SAGEConv** – Graph Neural Network Layers
 - **MeSH / UMLS / DO** – Medical Ontologies & Vocabularies
 - **SHAP / XAI** – SHapley Additive exPlanations & Explainable AI
 - **TxGNN** – Therapeutic target Graph Neural Network
 - **XGBoost** – Extreme Gradient Boosting
-

Chapter (1)

Introduction

1.1 Overview

Drug repurposing is the systematic process of identifying new therapeutic indications for existing, approved drugs leveraging already-validated compounds to reduce the cost, time, and clinical risk of bringing new treatments to patients. This project implements an advanced computational AI framework that addresses this challenge through a dual-model ensemble architecture grounded in biological knowledge graph data and machine learning.

The framework is built upon two complementary and independently trained models:

DREAMwalk (Drug Repurposing through Exploring Associations using Multi-layer random Walk) — a graph embedding pipeline operating on the Hetionet biological knowledge graph. It generates 128-dimensional node embeddings via semantic-guided random walks combining local traversal with similarity-based teleportation across compound and disease nodes. These embeddings feed an XGBoost classifier that predicts novel drug-disease associations, with SHAP explainability attributing predictions to specific pathways

TxGNN (Therapeutic target Graph Neural Network) — a deep graph neural network trained end-to-end on the same Hetionet heterogeneous graph using PyTorch Geometric. TxGNN employs a two-layer GraphSAGE message-passing encoder with relation-specific DistMult (Distributed Multiplicative) link scoring, learning drug and disease representations directly from the graph topology without relying on walk sequences. This architectural independence from DREAMwalk is what makes combining both models meaningful. The outputs of both models are unified through an **ensemble scoring layer** that produces a single, weighted confidence score for each drug-disease candidate pair. By requiring agreement between two architecturally different models — one based on random walk embeddings and gradient-boosted trees, the other on end-to-end neural message passing — the framework significantly reduces false positive predictions and improves robustness on novel, unseen associations

1.2 Motivation

The necessity for this project stems from the severe economic and temporal constraints imposed on modern pharmaceutical development:

- **High Costs and Prolonged Timelines:** Developing a completely new drug from scratch is an incredibly expensive process that takes over a decade of clinical and laboratory trials.
- **Advantages of Repurposing:** Leveraging existing, approved drugs significantly minimizes the risks of clinical failure related to toxicity. It also dramatically lowers overall development costs and shortens the regulatory approval pipeline (e.g., FDA approvals).
- **Emergency Health Response:** Global health crises, such as the COVID-19 pandemic, have highlighted the urgent need for rapid computational tools capable of finding immediate treatments from available stockpiles (e.g., discovering baricitinib or melatonin for new indications).
- **Shift Toward Data-Driven Discoveries:** With the massive influx of biomedical big data, there is a strong motivation to move away from narrow, hypothesis-driven models toward systematic, comprehensive data-driven exploration.

1.3 Challenges

1. Data Incompleteness

One of the primary challenges in biomedical knowledge graphs is data incompleteness. In Hetionet, only 755 out of 209,168 compound-disease pairs are known treatments, representing approximately 0.36% of the entire search space as shown in Figure 1.1. This extreme sparsity creates difficulties for machine learning models because most relationships remain unvalidated rather than truly negative. Consequently, drug repurpose models must perform link prediction under highly incomplete conditions, increasing uncertainty and reducing prediction reliability.

Both DREAMwalk and TxGNN address this directly through **negative sampling strategies** that carefully distinguish between truly absent associations and simply unvalidated ones — never treating an unlabeled pair as a confirmed negative during training.

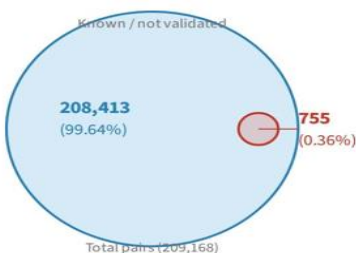


Figure 1. 1 shows Data Incompleteness

2. Data Heterogeneity

Biomedical knowledge graphs aggregate information from multiple heterogeneous resources as shown below in Figure 1.2 . Hetionet integrates 29 different databases with varying schemas, identifiers, and licensing conditions. Such heterogeneity complicates data integration and entity alignment processes. Inconsistent data representations may introduce noise, missing mappings, and semantic inconsistencies that negatively affect downstream machine learning performance.



Figure 1.0.2 shows Data Heterogeneity

3. Black Box Problem

Although the Rephetio framework evaluated 1,206 metapaths and identified 709 statistically significant patterns, only 31 features were retained in the final predictive model. Figure 1.3 shows us the distributions of the black box problem, the feature selection process lacks full interpretability, making it difficult to understand why specific biological pathways contribute to predictions. This limited transparency represents a major explainability challenge in AI-driven drug repurposing systems.

To overcome this limitation, our framework integrates Explainable AI (SHAP) and an interactive XAI-driven chatbot interface to transparently translate complex feature attributions and biological pathways into actionable, human-readable insights.

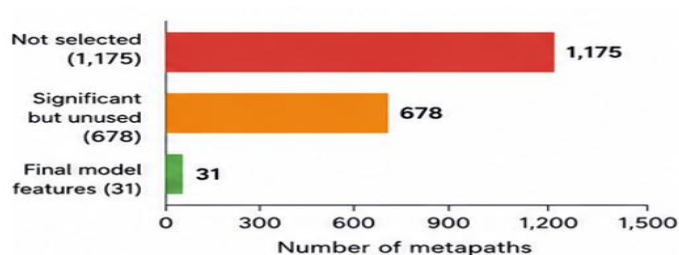


Figure 1.3 shows Black Box Problem

4. Generalizability Gap

The significant decline in AUROC across external validation datasets highlights a generalizability challenge. While the model achieves 97.4% AUROC on training indications, performance decreases to 70% when evaluated against clinical trial data. This gap as shown in Figure 1.4 suggests that the learned patterns may not fully transfer to unseen biomedical contexts, limiting real-world applicability.

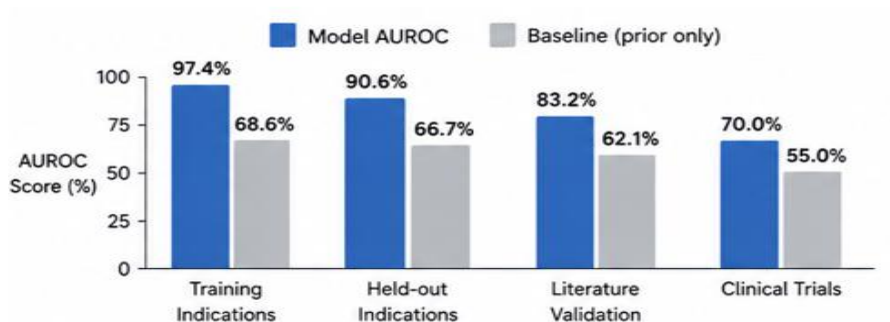


Figure 1.4 shows the Generalizability Gap

5. Clinical Risks

The epilepsy case study demonstrates how incomplete relationship information can lead to biologically implausible predictions. Figure 1.5 shows that Among the top 100 predicted compounds, 15 were already known to induce seizures. This limitation arises because the graph structure does not adequately represent directional pharmacological effects, preventing the model from distinguishing beneficial mechanisms from harmful ones.

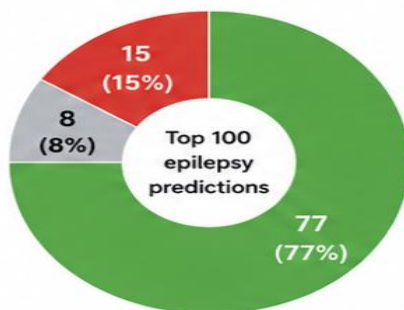


Figure 1. 5 shows clinical risk challenges

1.4 Project Objectives

The strategic and technical milestones of this project include:

- **Constructing a Balanced Unified Embedding Space:** Designing an algorithm that successfully balances biological pathways with semantic contexts, ensuring drugs and diseases are represented accurately without being overshadowed by dense gene data.
- **Developing Semantic Teleportation:** Implementing a mechanism that allows the random walk agent to dynamically hop across disconnected graph components based on calculated semantic similarity matrices, enriching the contextual data captured.
- **Maximizing Association Prediction Accuracy:** Outperforming existing state-of-the-art link prediction models across major standardized benchmarks (such as HetioNet, MSI, and KEGG).
- **Training an End-to-End Graph Neural Network:** Implementing TxGNN as a second, independent model that learns drug and disease representations directly from the Hetionet heterogeneous graph topology through multi-relational message passing and DistMult link scoring — without dependence on pre-generated walk sequences.
- **Building a Dual-Model Ensemble:** Combining DREAMwalk/XGBoost and TxGNN predictions through a weighted ensemble scoring layer, leveraging architectural diversity between the two models to produce more reliable drug-disease association scores than either model achieves individually.
- **Providing Biological Interpretability:** Ensuring the system acts as more than a "black box" by allowing researchers to trace the extracted random walk paths to better understand the proposed Mechanism of Action (MoA) behind a predicted drug-disease link.
- **Validation via Real-World Case Studies:** Demonstrating the practical efficacy of the model by generating valid, verifiable therapeutic hypotheses for complex conditions, specifically focusing on Breast Carcinoma and Alzheimer's Disease.
- **Developing an Interactive XAI Assistant:** Integrating an intelligent assistant powered by ensemble prediction scores and SHAP values to dynamically explain the evidence behind any predicted drug-disease association in human-readable terms for clinical end-users

This documentation provides a solid foundation for your project. We can proceed by expanding on the mathematical logic behind the semantic teleportation algorithm or detailing the specific dataset schemas whenever you are ready.

1.5 Thesis Organization

The remainder of this thesis is logically structured into the following chapters to present a comprehensive overview of the developed AI-driven drug repurposing framework:

- **Chapter 1: Introduction** — Introduces the research domain, establishes the project's core objectives, outlines the primary motivations regarding pharmaceutical constraints, and summarizes the technical challenges addressed by the proposed solution.
- **Chapter 2: Background and Literature Review** — Provides the essential biomedical and computational foundations, evaluates existing network-based drug repurposing models, and highlights the limitations of current state-of-the-art methods regarding gene dominance bias.
- **Chapter 3: Solution Approach** — Details the full architectural design of the framework, covering the DREAMwalk multi-layer graph construction, semantic-guided random walk mechanism, and embedding generation pipeline, followed by the TxGNN heterogeneous graph neural network architecture, DistMult link scoring, and the ensemble layer that combines both models
- **Chapter 4: Experimental Setup and Implementation** — Outlines the evaluation datasets utilized (Hetionet, KEGG), the configuration and hyperparameter tuning of both the XGBoost classifier and the TxGNN model, the leak-free cross-validation protocols applied independently to each model, and the SHAP-based explainability integration.
- **Chapter 5: Conclusion and Future Work** — Summarizes the core research findings from both models, emphasizes the practical contributions of the ensemble system and XAI interface, and discusses potential pathways for future enhancements including attention-based ensemble weighting, extended knowledge graph sources, and improved relation-aware scoring functions.

Chapter (2)

Background and Literature Review

2.1 Drug Structure

In computational drug discovery and drug repurposing, representing a drug's **molecular and chemical structure** is fundamental because molecular structure strongly influences its physicochemical properties, **molecular** interactions, pharmacological activity, and interactions with biological targets. However, molecular structure alone does not determine a drug's safety or toxicity, which can also depend on factors such as dose, exposure, metabolism, pharmacokinetics, and biological context.

SMILES (Simplified Molecular Input Line Entry System) is a line notation used to represent molecular structures as text strings. For example, ethanol can be represented as CCO [1].

Extended-Connectivity Fingerprints (ECFPs) are molecular fingerprints that encode local atom environments and molecular substructures and are widely used to calculate molecular similarity and support structure–activity modeling. In fixed-length implementations, the resulting features can be hashed into fingerprint positions represented as binary vectors [2].

In Hetionet, the **Compound–resembles–Compound (CrC)** relation represents chemical similarity between compounds. This relation is derived from molecular fingerprints and should be interpreted as a **chemical-similarity signal**, rather than evidence that structurally similar compounds necessarily have identical therapeutic effects or biological functions [3], [6].

In this project, the CrC relation is used as one source of structural information within the heterogeneous biomedical knowledge graph. It contributes to the representation of relationships between compounds and can support drug-candidate prioritization. **The specific use of CrC within the DREAMwalk and TxGNN-based branches should be described as part of the project implementation rather than automatically attributed to the original DREAMwalk or TxGNN publications.**

2.2 Drug Uses and Therapeutic Indications

Drug use refers to the therapeutic application of a medicinal substance. A therapeutic indication identifies the disease, medical condition, or therapeutic purpose for which a drug is used. An indication may be officially approved by a regulatory authority or may represent an off-label use. This distinction is particularly important in drug repurposing, where existing drugs are investigated for potential therapeutic applications beyond their established uses.

The Anatomical Therapeutic Chemical (ATC) classification system provides a standardized framework for classifying active substances according to the organ or system on which they act and according to their therapeutic, pharmacological, and chemical properties. The ATC/DDD methodology is maintained by the WHO Collaborating Centre for Drug Statistics Methodology and is used internationally for drug-utilization research [4].

In this project, drug-use information and ATC classifications provide semantic context that complements molecular structure and biomedical network relationships. Rather than representing a drug solely as a chemical compound, this information captures aspects of its therapeutic role and clinical context. This semantic information is particularly relevant to DREAMwalk, whose original methodology incorporates ATC-based drug semantic similarity into its semantic-information-guided random-walk process [5].

2.3 Biomedical Knowledge Graphs & Network Biology

Network biology represents biological entities as interconnected systems rather than isolated objects, enabling biological relationships to be analyzed as networks. In computational drug **repurpose**, these heterogeneous biomedical relationships can be represented as Biomedical Knowledge Graphs (BKGs), where nodes represent biomedical entities and typed edges represent relationships between them.

A BKG can contain heterogeneous entities such as compounds, genes, proteins, diseases, pathways, anatomical structures, and side effects, while its edges can represent relationships such as binding, interaction, association, treatment, or chemical similarity. This heterogeneous representation enables computational models to integrate different types of biomedical evidence within a unified graph.

Hetionet is a heterogeneous biomedical network specifically designed to support computational drug-repurposing research. Hetionet v1.0 contains 47,031 nodes, 24 relation types, and 2,250,197 relationships. Relevant relation types include Compound–binds–Gene, Compound–resembles–Compound, Compound–treats–Disease, Disease–associates–Gene, and Gene–interacts–Gene [3].

A key principle used in network-based biological inference is Guilt-by-Association (GBA). GBA assumes that information about the function or characteristics of a biological entity can be inferred from its relationships and network context, particularly when it is connected to entities with known properties [7]. In drug repurposing, this principle can be used to infer potential drug–disease associations from interconnected drug, gene, and disease relationships.

DREAMwalk extends this network-based reasoning to a semantic multi-layer setting by integrating drug, gene, and disease information and using semantic-information-guided teleportation during random walks. The original study identifies the dominance of dense

gene–gene networks as a potential source of bias and introduces semantic teleportation to improve drug and disease representation learning [5].

In the present project, the literature concepts are distinguished from the project implementation. The proposed system combines a DREAMwalk/XGBoost branch, a TxGNN-based graph-learning branch, and an ensemble layer. These integration choices represent the architecture implemented in this project and should not be interpreted as components of the original DREAMwalk or TxGNN publications.

2.4 Related Work

Table 2.1 separates the six distinct works represented by the supplied chapter. TxGNN and deepDR are kept as separate references because they are different publications with different datasets, architectures, and evaluation protocols. Numerical results are included only where they are supported by the corresponding source.

Table 2.1 summarizes the four research papers reviewed in this section, including their authors, titles, methodologies, datasets, results, and proposed future work.

NO.	WORK YEAR	/ CORE METHOD	DATA EVALUATION	/ REPORTED RESULTS	LIMITATIONS DIRECTION	/ FUTURE
1	Bang et al. (2023) – DREAMwalk [5]	Semantic information guided multi-layer random walks; heterogeneous Skip-gram representation learning; XGBoost prediction	MSI, HetioNet, and KEGG; evaluated using Accuracy, AUROC, and AUPR	Average Accuracy = 0.873; AUROC = 0.938; AUPR = 0.939; improvement of up to 16.8% over comparison methods	Addresses the influence of highly connected gene/PPI regions through semantic-information-guided teleportation.	
2	Huang et al. (2024) – TxGNN [8]	Graph foundation model for drug repurposing; heterogeneous GNN with metric learning; indication and contraindication prediction; TxGNN Explainer	PrimeKG; 17,080 diseases and 7,957 drugs; stringent zero-shot evaluation	Reported improvements of 49.2% for indication prediction and 35.1% for contraindication prediction versus benchmark methods under the reported zero-shot evaluation	Focuses on zero-shot generalization and interpretable multi-hop reasoning. The original work is based on PrimeKG, not Hetionet.	
3	Zeng et al. (2019) – deepDR [9]	Random-walk-based representation of multiple drug-related networks, followed by	10 biological networks; 1,519 approved drugs, 1,229 disease terms, and 6,677 clinically reported	Cross-validation AUROC = 0.908; AUPR = 0.923; external validation AUROC = 0.826	Performance depends on the completeness and quality of available biological associations and integrated networks.	

		multimodal deep autoencoder (MDA) and collective variational autoencoder (cVAE) learning.	drug–disease pairs; 5-fold CV repeated 10 times; ClinicalTrials.gov external validation		
4	Wang et al. (2022) – DrugRepo [10]	Explicit scoring framework using structural similarity (TC), overlapping compound–target score (OCTS), and compound–disease score (CDS)	Chemical structures, drug–target interactions, pathways, and disease–gene associations; 669 diseases across 22 disease groups	At DrugRepo score ≥ 0.4 : 516 approved drugs, 545 diseases, and 6,229 drug–disease pairs	Depends on available drug–target and related biological evidence; authors suggest incorporating additional data types.
5	Amiri et al. (2023) – IDDI-DNN [11]	Three drug–similarity matrices and two disease–similarity matrices integrated using Similarity Network Fusion, followed by CNN-based prediction	Gold-standard and DNdataset datasets; ROC and precision–recall evaluation	Reported to outperform comparison methods on the evaluated datasets; exact numerical values should be taken directly from the original paper where required	Authors highlight the need for broader biological information, larger and more diverse datasets, and improved generalizability.
6	Talevi & Bellera (2020) [12]	Review of drug–repurposing strategies, challenges, opportunities, and publication trends; not an ML prediction model	Literature review and Scopus publication-trend analysis	Identifies scientific, regulatory, economic, intellectual-property, data-integration, and translational challenges	Used as strategic/background literature rather than as a predictive-model benchmark.

2.4.1 Synthesis and Positioning of This Work

The comparative analysis of the six reviewed works demonstrates a progression from similarity- and network-based drug-repurposing approaches toward deep representation learning, heterogeneous graph learning, and graph foundation models. As summarized in Table X, the reviewed studies employ different data sources, representations, modeling assumptions, and evaluation protocols. Their contributions and limitations are therefore complementary rather than identical, providing several directions that motivate the design of the present work.

Heterogeneous evidence integration. The reviewed literature employs multiple strategies for representing relationships between drugs, diseases, and biological entities. DREAMwalk uses semantic-information-guided random walks over heterogeneous biomedical networks; deepDR integrates multiple biological networks through deep representation learning; DrugRepo combines chemical and genomic evidence through explicit scoring components; IDDI-DNN integrates multiple drug and disease similarity matrices; and TxGNN formulates drug repurposing as a heterogeneous graph-learning problem with zero-shot prediction capability. These approaches demonstrate the value of combining heterogeneous biomedical evidence while also highlighting that different representations capture different aspects of drug–disease relationships.

Model diversity and ensemble motivation. The proposed combination of two modeling branches is a project-level design choice motivated by their complementary representations and inductive biases. The present work combines a DREAMwalk/XGBoost branch, which provides network-based semantic representations, with a TxGNN-based heterogeneous graph-learning branch, which models typed biomedical relationships. Their predictions are subsequently integrated through an ensemble mechanism. This design is intended to investigate whether complementary predictive signals can improve drug-repurposing prioritization compared with relying on a single modeling perspective. The combination proposed in this thesis should be understood as an original project-level integration, rather than as a system introduced by the original DREAMwalk or TxGNN publications.

Integrated explainability. The reviewed studies provide different forms of interpretability. DREAMwalk provides sampled biological paths and downstream biological enrichment that can support interpretation, while TxGNN incorporates a dedicated Explainer for identifying multi-hop knowledge-graph reasoning. DrugRepo also provides interpretable component-level scores through its scoring framework. However, these mechanisms are specific to their respective methodologies and are not necessarily presented as a unified prediction and user-facing explanation workflow. The present work therefore incorporates SHAP-based feature attribution and biological interpretation as an integrated explainability layer accompanying the prediction process.

Network-structure considerations. Network structure is particularly relevant to random-walk-based biomedical representation learning. DREAMwalk identifies the high connectivity and density of gene/PPI regions as a potential source of bias in network-based representation learning and addresses this issue through semantic-information-guided teleportation, which increases transitions toward semantically related drug and disease nodes. The proposed framework complements this representation with a heterogeneous graph-learning branch that explicitly models different types of biomedical relationships. This complementary design does not assume that the graph-learning branch eliminates network-structure bias; rather, it provides an alternative representation of the same biomedical domain.

Positioning of the present work. Based on the reviewed literature, the proposed system is positioned as an ensemble-oriented drug-repurposing framework that integrates a DREAMwalk/XGBoost branch and a TxGNN-based heterogeneous graph-learning branch over the selected biomedical knowledge-graph data, followed by ensemble prediction and an integrated explainability layer. The objective is to investigate whether combining complementary network-based and heterogeneous graph-based representations can improve drug-repurposing prioritization while providing interpretable evidence for the resulting predictions. Importantly, all performance measurements, ensemble results, and explainability outputs produced by this integrated framework are results of the present project and should be clearly distinguished from the results reported in the original publications.

Chapter (3)

Solution Approach

3.1 Overview of DREAMwalk (XG-Boost)

DREAMwalk (Drug Repurposing through Exploring Associations using multi-layer random walk) is a computational framework for drug repurposing that operates on heterogeneous biomedical knowledge graphs (biomedKGs). Proposed by Bang et al. (2023) in Nature Communications, the model addresses a fundamental challenge in biomedKG-based learning: the structural dominance of gene-gene (protein-protein interaction; PPI) networks over drug and disease nodes, which leads to biased and less effective node representations for the entities most critical to drug repurposing.

The core innovation of DREAMwalk is the introduction of a semantic multi-layer guilt-by-association (GBA) principle. In classical network biology, GBA posits that entities sharing network neighbors tend to share biological functions. DREAMwalk extends this idea to a multi-layer context: drugs and diseases are not only understood through their molecular-level neighbors (target proteins, pathway genes) but also through their semantic neighbors — other drugs or diseases with analogous therapeutic or nosological classifications. This dual perspective allows the model to generate embedding spaces where therapeutically related entities cluster together, even when their direct molecular interactions diverge.

3.2 Architecture and Pipeline of XG-Boost

3.2.1 Stage 1: Optimized Data Ingestion

- Figure 3.1 shows a Comprehensive Hetionet v1.0 Architecture: Hetionet v1.0 is an integrative heterogeneous network (hetnet) constructed from 29 public biomedical resources, designed to unite diverse biological domains into a single graph structure in which nodes represent distinct biological entities (nouns), and edges represent experimental or curated relationships (verbs). In total, the network encodes:
 - 47,031 Nodes spanning 11 Metanode Types

- 2,250,197 Relationships spanning 24 Metaedge Types

1. Metanodes (Node Types & Counts)

The network structure incorporates 11 foundational biological categories:

- Gene (G): 20,945 protein-coding human genes
- Biological Process (BP): 11,381 Gene Ontology terms
- Side Effect (SE): 5,734 UMLS-coded drug side effects (from SIDER)
- Molecular Function (MF): 2,884 Gene Ontology terms
- Compound (C): 1,552 approved small-molecule drugs (from DrugBank)
- Cellular Component (CC): 1,391 Gene Ontology terms
- Pathway (PW): 1,822 integrated biological pathways (WikiPathways, Reactome, PID)
- Anatomy (A): 402 anatomical structures/tissues (from Uberon)
- Symptom (S): 438 clinical signs and symptoms (from MeSH)
- Pharmacologic Class (PC): 345 FDA classification categories (from DrugCentral)
- Disease (D): 137 standardized complex diseases (DO Slim)

2. Metaedges (Relationship Types)

Edges represent multiplex connectivity across these domains, encompassing low-throughput experimental binding, high-throughput omics signatures, genetic associations, and clinical indications. Key metaedge categories include:

- Compound-Target & Pharmacology: Compound-binds-Gene (CbG), Compound-treats-Disease (CtD), Compound-palliates-Disease (CpD), and Pharmacologic Class-includes-Compound (PCIC).
- Genetics & Molecular Interaction: Gene-interacts-Gene (GiG), Disease-associates-Gene (DaG), and Gene participation terms across pathways, biological processes, molecular functions, and cellular components.
- Phenotypic & Clinical Observations: Compound-causes-Side Effect (CCSE), Disease-presents-Symptom (DpS), Disease-localizes-Anatomy (DlA), and chemical/disease similarity mappings (CrC, DrD).
- Transcriptional Profiles: Regulatory and expression edges derived from LINCS L1000, Bgee, and STARGEO meta-analyses (CdG, CuG, AdG, AuG, AeG, DdG, DuG).

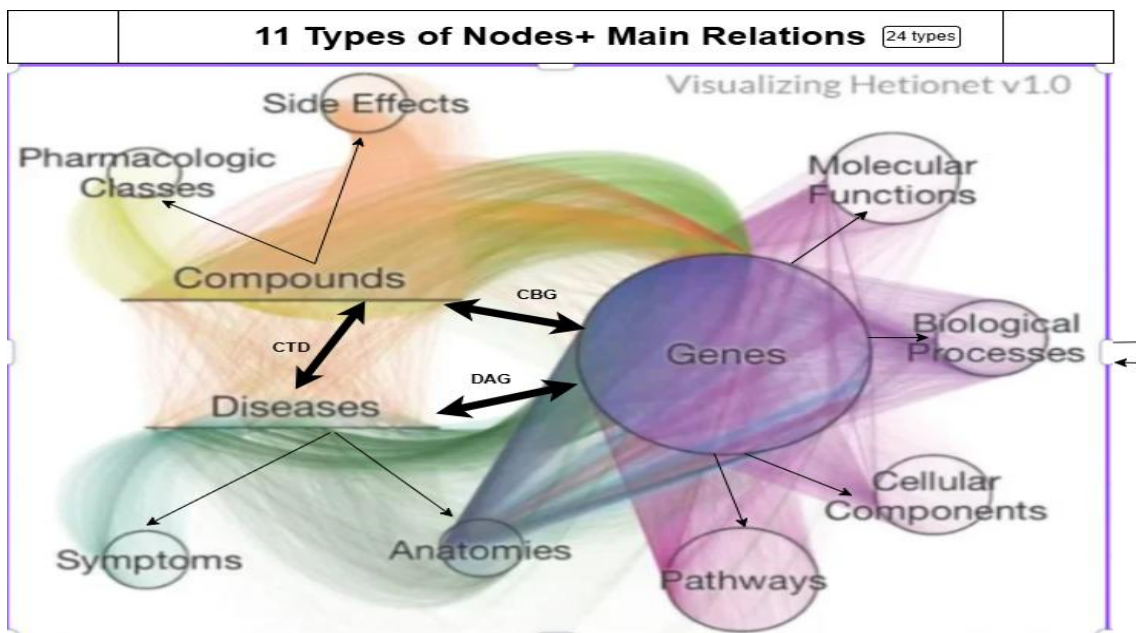


Figure 3.1 Hetionet v1.0 Network structure and Relations

3.2.2 Stage 2: Leakage-Free Cross-Validation Protocol

- **Rigorous Isolation & Prevention:** A core architectural principle of this pipeline is the strict enforcement of a leakage-free evaluation framework. In naive graph embedding implementations, data leakage occurs when random walks and similarity matrices are computed globally, implicitly exposing test-set relationships to the training environment.
- **Dynamic Fold Reconstruction:** Throughout the ten-fold stratified cross-validation process, all test-fold Compound-treats-Disease (CtD) edges are dynamically stripped from a deep copy of the graph (G_{train}) before the generation of any structural similarity matrices or random walks.
- **Absolute Experimental Integrity:** Consequently, similarity matrices and vector embeddings are completely rebuilt from scratch using exclusively G_{train} at every single cross-validation fold, ensuring absolute experimental integrity and unbiased model evaluation.

In this implementation, leakage is prevented by removing all test-fold CtD edges from the graph before the random walk phase in each cross-validation iteration. Specifically, the execution sequence within each fold is summarized in **Table 3.1**.

Table 3.1 summarizes Step-by-Step Cross-Validation Protocol and Pipeline Execution

Step	Action
1	Split CtD pairs into train and test folds (stratified) (Data Integration)
2	Remove test-fold CtD edges from the HetioNet graph (Leakage-Free Protocol)
3	Sparse Graph Representation
4	Vectorized Jaccard Similarity
5	Run teleport-guided random walks on the reduced graph
6	Train Word2Vec (Skip-gram) on the generated node sequences
7	Extract drug/disease embeddings; construct feature vectors
8	Train XGBoost classifier on train-fold feature vectors
9	Evaluate on held-out test-fold; compute AUROC, AUPR, Accuracy

This protocol ensures that the reported performance metrics reflect the model's true generalization capability, making the evaluation more conservative and more representative of real-world drug repurposing scenarios where test associations are completely unseen during training.

3.2.3 Stage 3: Sparse Graph Representation

To avoid the high memory overhead and slow iteration times of standard dictionary lookups (such as NetworkX), the system constructs high-performance Compressed Sparse Row (CSR) matrices using SciPy and NumPy. This optimizes bidirectional edge management, neighbor precomputation, and vector-level calculations across massive heterogeneous structures.

- **Execution Mode:** The construction operates partially in parallel via **Vectorized Construction**.
- **Detailed Mechanism:** Building sparse matrices (CSR / CSC) avoids slow, conventional loops that iterate over individual edges. Instead, operations are executed concurrently in a vectorized manner using specialized libraries such as SciPy.
- **Memory-Level Parallelism:** The underlying backend distributes the construction of row and column indices as well as data arrays across memory all at once, achieving efficient memory-level parallelism. The conversion of the network topology into an efficient adjacency matrix format is shown in **Figure 3.2**.

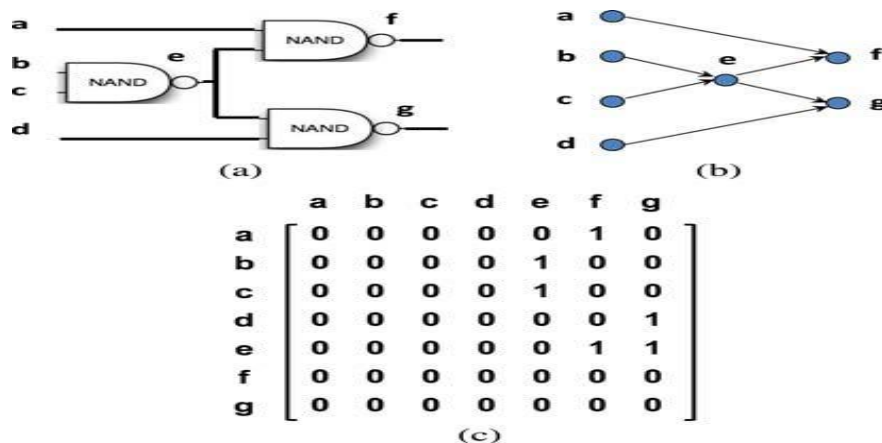


Figure 3.2 shows Transformation of network topology into a sparse adjacency matrix

3.2.4 Stage 4: Vectorized Jaccard Similarity

- **Efficient Metric Computation:** To quantify structural neighborhood overlaps across massive biomedical nodes without performance bottlenecks, the system computes Jaccard similarity using fully vectorized matrix operations over sparse representations.
- **Matrix-Level Acceleration:** By replacing slow iterative loops with optimized sparse array dot products and broadcasting operations, structural similarities between entity neighborhoods are calculated concurrently.
- **Downstream Integration:** The resulting similarity matrices feed directly into subsequent pipeline stages, providing robust structural relationship metrics to enhance embedding quality and link prediction accuracy, as illustrated in **Figure 3.3**.

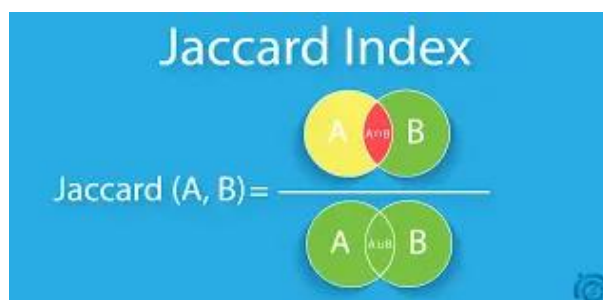


Figure 3.3 shows Jaccard Index Mathematical Formula and set operations

3.2.5 Stage 5: Semantic Information-Guided Teleport Random Walk

The first stage generates informative node sequences from the biomedKG. A random walker traverses the graph following edge weights modulated by an edge-type transition matrix (inspired by edge2vec). When the walker arrives at a drug or disease node, it probabilistically selects between two actions:

- **Network Traversal** (probability $1 - \tau$): The walker proceeds along the graph topology as usual.

- **Teleport Operation (probability τ):** The walker jumps to a semantically similar entity, sampled from a precomputed drug-drug or disease-disease semantic similarity matrix S .

Semantic similarity is computed using an information-content (IC) based measure derived from established biomedical ontologies. For drugs, the Anatomical Therapeutic Chemical (ATC) classification hierarchy is used; for diseases, Medical Subject Headings (MeSH), Disease Ontology, or ICD-11 hierarchies are employed depending on the biomedKG. The IC-based similarity measure (Jiang & Conrath, 1997) is:

$$\text{sim}(c_1, c_2) = 1 - [\text{IC}(c_1) + \text{IC}(c_2) - 2 \cdot \text{IC}(\text{MICA}(c_1, c_2))] / 2$$

where MICA denotes the Most Informative Common Ancestor in the ontology hierarchy. A user-defined similarity cutoff (default: 0.4) is applied to reduce noise and computational overhead. The teleport factor τ controls the balance between biological and semantic information in the generated paths. For each node, 100 walks of length 4 are sampled (in the original paper; adapted in this implementation as described in Section 3)

3.2.6 Stage 6: Heterogeneous Skip-gram Node Embedding

- **Node Representation Learning:** The teleport-guided random walk sequences are passed to a heterogeneous Skip-gram model for node representation learning. Unlike standard node2vec, the heterogeneous Skip-gram performs negative sampling exclusively within the same node type, producing type-aware embedding vectors that respect the heterogeneous structure of the biomedKG. Inspired by metapath2vec++, this step generates a unified embedding space where drug and disease nodes can be meaningfully compared.
- **Feature Vector Construction:** For each drug-disease pair, a feature vector is constructed by element-wise subtraction of the two entities embedding vectors. An XGBoost classifier is trained on labeled drug-disease association (DDA) pairs to predict treatment probability. For drug repurposing, all unlabeled drug-disease pairs are scored, and candidates are ranked by their predicted association probability.

The structural overview of the Skip-gram architecture and the resulting embedding space are shown in **Figure 3.4**.

- **biomedKG $\rightarrow$ Teleport-guided Random Walks $\rightarrow$ Heterogeneous Skip-gram $\rightarrow$ Node Embeddings $\rightarrow$ XGBoost Classifier $\rightarrow$ DDA Predictions**

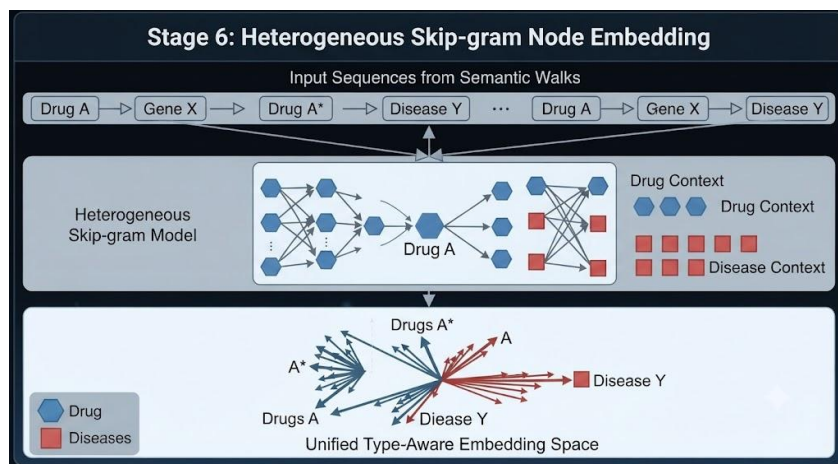


Figure 3.4 Heterogeneous Skip-gram Model Architecture and Unified Embedding Space

3.2.7 Stage 7: Supervised Classification via XGBoost

- **Model Optimization:** An XGBoost classifier handles the final supervised link prediction task. The model is configured with hyperparameters including 300 estimators, a learning rate of 0.05, max tree depth of 6, and histogram-based training (tree_method = "hist").
- **Early Stopping & Training:** Cross-validation folds incorporate evaluation monitoring and early stopping, followed by a final production model trained on the full graph dataset to generate global prediction scores (pred_df). The sequential architecture of the XGBoost ensemble mechanism is illustrated in Figure 3.5.

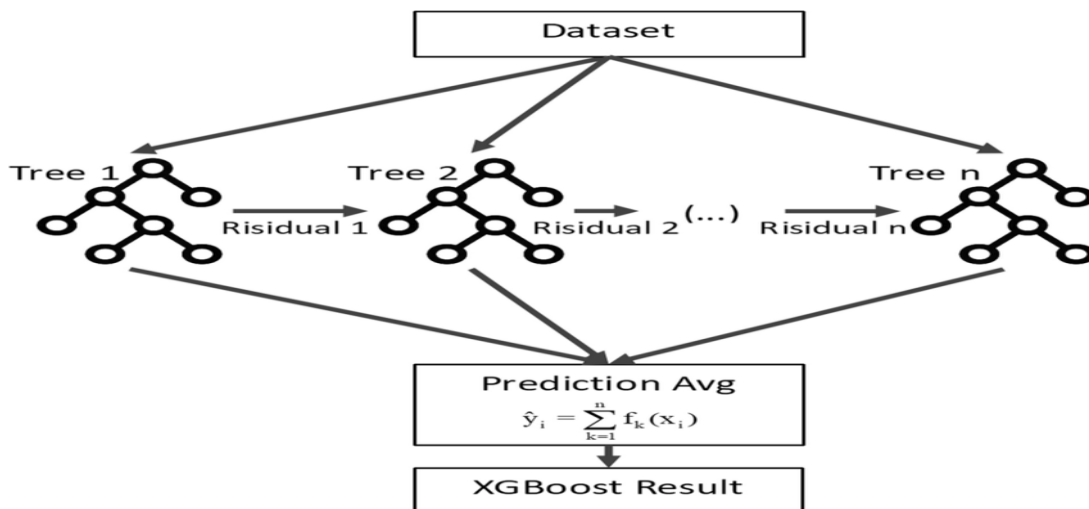


Figure 3.5 Architectural Overview of the XGBoost Ensemble Process

3.2.8 Stage 8: Explainable AI (XAI) Suite & Inference

To ensure human-centered interpretability, the framework integrates an advanced XAI suite powered by **SHAP (SHapley Additive Explanations)** values and feature attribution

layers. This parse background datasets mapped directly to real drug and disease names to explain model confidence scores and decision pathways.

Global vs. Local Interpretability:

- **Global Interpretability:** Evaluates the overarching dataset behavior, highlighting the **main global features** that consistently drive the model's predictions across all biomedical entities, helping identify general patterns and high-impact biological markers.
- **Local Interpretability:** Focuses on individual predictions, allowing the system to isolate a **specific drug or disease** prediction and extract the exact local features and contributions that led the model to recommend or link that particular entity.

3.3 Overview TxGNN Foundation Model

Before presenting the implementation developed in this project, this section summarizes the original TxGNN model that motivated it. TxGNN is a graph foundation model for drug repurposing, introduced by Huang et al. [1] and trained on a large-scale biomedical knowledge graph to predict drug indications and contraindications across thousands of diseases. The figures reproduced in this section are adapted from the original publication for background and comparison purposes only; they do not depict outputs of the implementation described in Section 3.2.

The overarching concepts, motivation, and scale of the underlying knowledge graph used in TxGNN are presented in **Figure 3.6**.

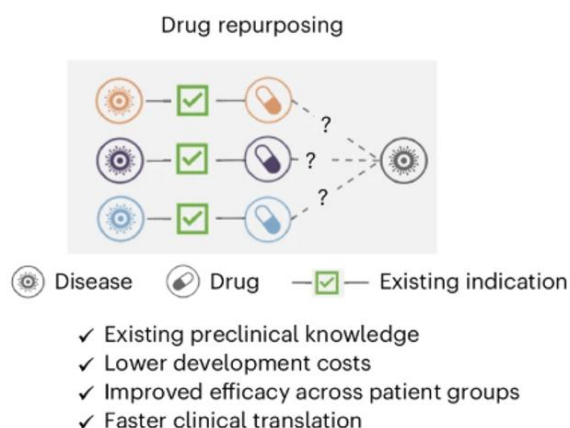


Figure 3.6 Concept of Drug Repurposing and Clinical Advantages (Adapted from Huang et al. [1]).

Furthermore, the core architecture of TxGNN, including its predictor and explainer modules, is depicted in **Figure 3.7**.

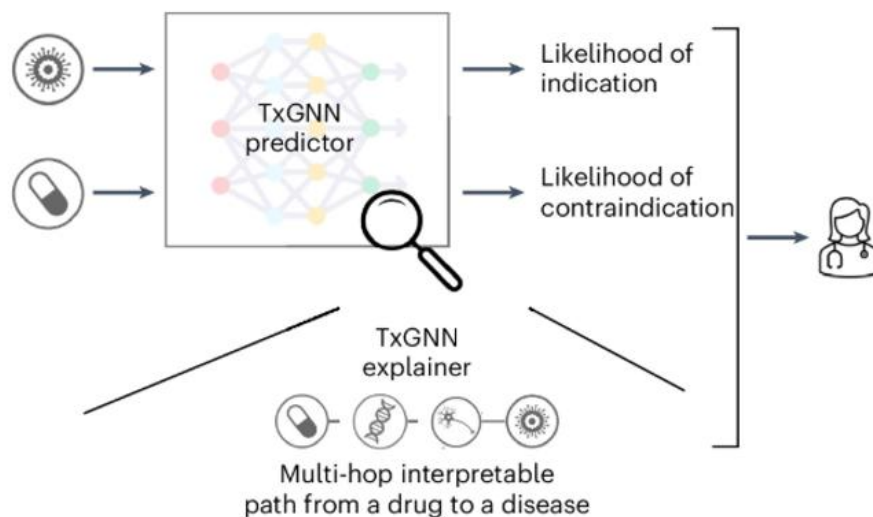


Figure 3.7 Overview of the TXGNN Predictor and Multi-hop Explainer Modules (Adapted from Huang et al. [1])

Finally, the scale and entity diversity of the underlying biomedical knowledge graph used to train the model are summarized in **Figure 3.8**.

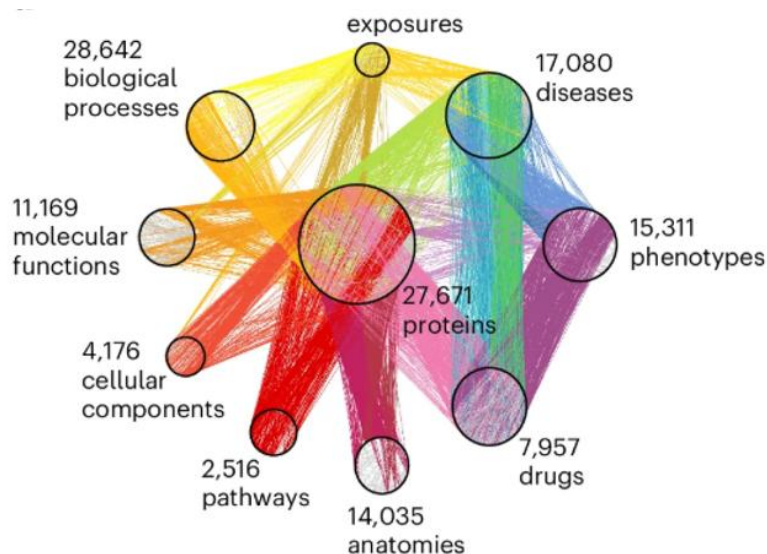


Figure 3.8 Biomedical Knowledge Graph Scale and Heterogeneous Entity Types (Adapted from Huang et al. [1])

Furthermore, the graph neural network message-passing mechanism and metric-learning logic are illustrated in **Figure 3.9**.

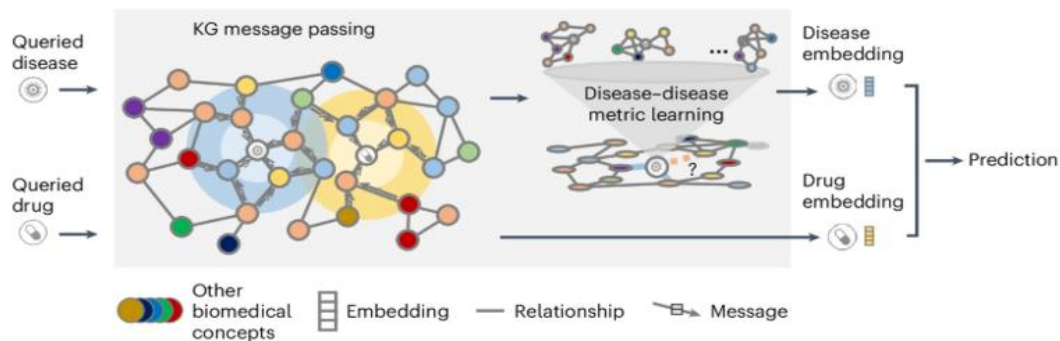


Figure 3.9 Knowledge Graph Message-Passing and Metric-Learning Architecture (Adapted from Huang et al. [1])

The distribution of pairwise disease-signature similarity scores across the knowledge graph is presented in **Figure 3.10**.

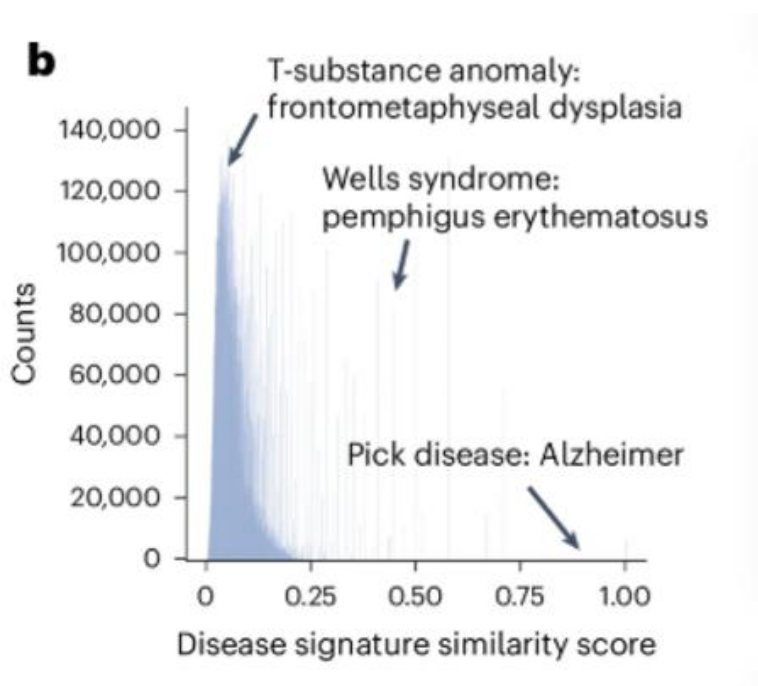


Figure 3.10 Pairwise Disease-Signature Similarity Score Distribution (Adapted from Huang et al. [1])

Finally, the predictive evaluation performance of TxGNN compared to baseline models under a random disease split is summarized in **Figure 3.11**.

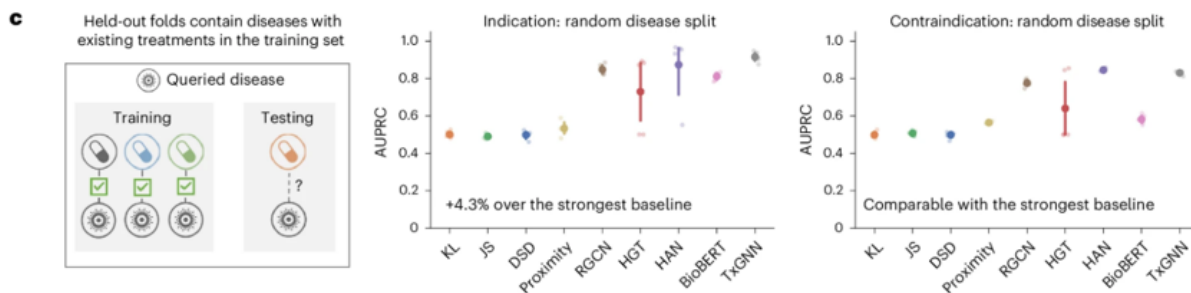


Figure 3.11 Benchmarking Performance Comparison Under Random Disease Split (Adapted from Huang et al. [1])

3.4 Architecture and Pipeline TXGNN

3.4.1 Stage 1: Local Data Ingestion

The pipeline begins by loading the Hetionet v1.0 knowledge graph directly from local Parquet files (nodes.parquet and edges.parquet) using Pandas, rather than downloading or re-parsing the original TSV distribution at runtime. This loading step is deliberately executed as the very first operation in the notebook, before torch or torch_geometric is imported. This ordering ensures that Pandas and PyArrow have full memory headroom while parsing the graph and avoids any shared-memory contention with a CUDA context that has not yet been created.

Only the columns actually consumed by the downstream pipeline are read (id, name, kind for nodes; source, metaedge, target for edges), which avoids materializing unused columns present in the Parquet files. The kind column of the node table and the metaedge column of the edge table — each of which takes only a small number of distinct values — are cast to the Pandas category dtype, which substantially reduces memory footprint while leaving all subsequent .value_counts(), .isin(), and dictionary-construction operations functionally unchanged. A dedicated Compound-treats-Disease (CtD) subset is also extracted at load time for early inspection.

In parallel, a single local output directory is created to persist every checkpoint, table, and figure the notebook produces later, replacing the ephemeral Colab/Drive paths used in earlier iterations of the project. The output of this stage is a pair of in-memory DataFrames (df_nodes, df_edges) together with a resolved local OUTPUT_DIR, which feeds directly into the indexing stage that follows. An environment variable workaround (KMP_DUPLICATE_LIB_OK) is also set at this point to prevent a common OpenMP runtime conflict between PyTorch and PyArrow on local Windows environments.

3.4.2 Stage 2: Environment and Dependency Setup

Once the raw data is safely loaded, the notebook proceeds to import PyTorch, PyTorch Geometric, and the supporting scientific-computing stack (NumPy, scikit-learn, Matplotlib, tqdm). PyTorch Geometric's HeteroData container and its heterogeneous convolution layers (HeteroConv, SAGEConv, GATConv, RGCNConv, HGTCConv,

Linear) are imported for later use, although the final architecture relies specifically on HeteroConv combined with SAGEConv.

A fixed random seed (`torch.manual_seed(42)`) is set to make experiments reproducible, and the compute device is resolved dynamically (cuda if a GPU is available, otherwise cpu), with the active device name printed for confirmation. This stage produces the shared device object and confirms library versions before any graph construction begins.

3.4.3 Stage 3: Node Indexing and Knowledge-Graph DataFrame Construction

To translate the raw Hetionet tables into a form usable by a heterogeneous GNN, the pipeline builds a set of lookup structures over the node table: a mapping from node ID to node type (`id_to_type`), a mapping from node ID to human-readable name (`id_to_name`), and a per-type list of node IDs (`node_type_to_nodes`). From these, a global (`node_type, node_id`) $\rightarrow$ local_index dictionary (`node_to_idx`) is constructed, giving every node a zero-based positional index within its own type — a requirement of PyG's HeteroData, which indexes nodes per type rather than globally.

The two drug-disease relation codes used throughout the project, Compound-treats-Disease (CtD, the indication relation) and Compound-palliates-Disease (CpD, the palliation relation), are defined as named constants at this point. The full edge table is then iterated to build a unified knowledge-graph DataFrame, `df_kg`, in which every edge is resolved to its source type, source local index, relation name, destination type, and destination local index; edges whose endpoints cannot be resolved to a known type or index are dropped. `df_kg` is the central tabular representation of the graph used by every later stage, from splitting to negative sampling to the explainability module.

3.4.4 Stage 4: Leakage-Free Drug–Disease Split

Because the objective of the project is to predict unseen Compound-treats-Disease associations, the pipeline isolates all CtD/CpD edges from the rest of the knowledge graph (`df_dd` versus `df_other`) before any graph is built. `df_dd` is split into training, validation, and test partitions using scikit-learn's `train_test_split` with a fixed random9 state: 70% of the drug-disease pairs are retained for training, and the remaining 30% is further divided so that roughly 9.9% of the original pairs form the validation set and roughly 20.1% form the held-out test set.

Critically, the graph used for message passing during training, `df_train_full`, is assembled only from the non-drug-disease edges (`df_other`) plus the training-fold CtD/CpD edges (`df_train_dd`); validation and test associations are never inserted into the graph the model propagates over. A set of known training-fold positive pairs, `known_positive_pairs`, is also recorded at this stage and reused later to guarantee that negative samples never accidentally coincide with a true training association. This mirrors the leakage-prevention principle common to knowledge-graph link-prediction pipelines: structural information about held-out associations must never reach the model, directly or indirectly, during training.

3.4.5 Stage 5: Heterogeneous Graph Construction (PyTorch Geometric HeteroData)

The leakage-free training DataFrame, `df_train_full`, is converted into a PyTorch Geometric (PYG) HeteroData object through a dedicated `build_hetero_graph` routine. For every node type present in `node_type_to_nodes`, the corresponding node count is registered on the graph object. Edges are then grouped by canonical relation triple (`source_type`, `relation`, `destination_type`) and collected into per-relation source/destination index lists using a `defaultdict`, after which each group is converted into a single PyG `edge_index` tensor.

For every forward relation encoded in the data, an explicit reverse relation is also added under a `rev_` prefix, so that information can propagate in both directions during message passing even though Hetionet edges are stored in only one direction. The resulting graph exposes a full catalogue of node types and canonical edge types, and an index dictionary (relations) mapping each canonical edge type to an integer ID is derived from it for later use inside the model's relation-specific parameters. This heterogeneous graph, moved to the active compute device as `G_dev`, is the structure over which every subsequent forward pass of the model operates.

3.4.6 Stage 6: TxGNN Model Architecture

The core predictive model, implemented as the `TxGNN_PyG` class, follows an encoder–decoder design typical of knowledge-graph embedding models, adapted to a heterogeneous graph neural encoder. In the encoder, every node type is assigned its own learnable embedding table (`nn.Embedding`, Xavier-uniform initialized), of dimensionality equal to the input size, which serves as the initial node feature since Hetionet nodes carry no native features.

These embeddings are passed through two stacked heterogeneous convolution blocks built with PyG's `HeteroConv`, each wrapping a `SAGEConv` (GraphSAGE-style, mean-aggregating) layer per canonical edge type; the outputs of the different relation-specific convolutions incident on a node type are combined via sum aggregation. A ReLU non-linearity is applied after the first convolutional layer, and node types that receive no incoming messages at a given layer simply retain their pre-layer representation, guaranteeing that every node type always has a valid embedding regardless of local graph connectivity. The result of the second convolution layer is the final set of type-indexed node embeddings returned by `get_embeddings`.

In the decoder, link scores are computed with a `DistMult`-style bilinear form: each relation is assigned its own learnable weight vector (Xavier-initialized, one per canonical edge type, with / characters in relation names sanitized for use as PyTorch module-dictionary keys), and the score for a candidate edge is the element-wise product of the source embedding, the relation weight vector, and the destination embedding, summed over the embedding dimension. A learnable scalar temperature parameter is also defined on the model, initialized to 1.0.

The predict method resolves the two node types and the relation from a canonical edge-type triple and applies the DistMult scoring function, while forward computes matching positive and negative scores in a single call for use in the training loss. This architecture is deliberately parameter-efficient relative to the size of Hetionet: rather than learning a separate weight matrix per relation for message passing at every layer (as in a full RGCN), representation learning is delegated to shared SAGEConv layers, and relation-specific behaviour is concentrated in the lightweight DistMult decoder.

3.4.7 Stage 7: Leakage-Aware Negative Sampling

Because Hetionet, like most biomedical knowledge graphs, encodes only positive (observed) relationships, negative examples must be generated synthetically for supervised link prediction. The pipeline implements this with a `negative_sampling` function that performs rejection sampling: drug and disease indices are drawn uniformly at random (via NumPy's `default_rng` with a fixed seed) and a candidate pair is accepted only if it does not already appear in the relation's set of known positive pairs; sampling continues until the requested number of negatives — by default matching the number of positives, i.e. a 1:1 ratio — has been collected.

To support this efficiently across all relations, not just CtD, a dictionary of known positive pairs per relation, `all_positive_pairs`, is precomputed once from `df_train_full` prior to training. This same rejection-sampling routine is reused unmodified across the pretraining loop, the fine-tuning loop, the validation/test evaluation functions, and the baseline models, ensuring that the definition of a "negative" is consistent everywhere in the pipeline and that no evaluation negative can coincide with a genuine training-fold association.

3.4.8 Stage 8: Two-Phase Training Strategy

Model optimization proceeds through a shared `run_epoch` routine that, for each relation type in a supplied relation list, mini-batches the corresponding positive edges (batch size 512), draws matching negatives via the Stage 7 sampler, runs a forward pass of TxGNN_PyG to obtain positive and negative DistMult scores, concatenates them with binary labels, and optimizes a binary cross-entropy loss (`BCEWithLogitsLoss`) with an AdamW optimizer.

Training is organized into two explicit phases rather than a single monolithic loop. In Phase 1 (pretraining), the model is trained for 10 epochs over every relation type present in `df_train_full` — not just CtD — with a learning rate of $1e-3$ and weight decay of $1e-5$, so that the shared graph encoder learns general-purpose, structurally informed node embeddings from the full multi-relational context of Hetionet before it is specialized to the target task.

In Phase 2 (fine-tuning), training continues for 20 further epochs restricted to the CtD relation alone, with the learning rate lowered to $1e-4$ to adapt the model gently toward the drug-repurposing objective without disrupting the pretrained representations. During fine-tuning, the model is evaluated on the validation drug-disease split after every epoch using

the AUPRC metric, and the parameter state achieving the best validation AUPRC is checkpointed in memory (`best_state`) and restored at the end of training, giving early-stopping-like protection against overfitting on the small CtD fine-tuning set. The final selected weights are subsequently persisted to disk together with the associated best validation AUPRC and test-set AUROC/AUPRC.

3.4.9 Stage 9: Evaluation Pipeline

Model quality is assessed with a dedicated evaluate function that regenerates node embeddings from the current model state, retrieves the positive drug–disease edges for a given split, draws matching negatives through the Stage 7 sampler while explicitly excluding all known positive pairs for that relation, scores both sets with the model's DistMult decoder, and reports discrimination performance using ROC-AUC and average precision (AUPRC) via scikit-learn.

Beyond this ranking-based evaluation, a complementary set of functions (`evaluate_accuracy`, `evaluate_all_metrics`, `test_baseline_all_metrics`) converts the same positive/negative score pairs into hard classification outcomes at a fixed decision threshold of 0.5, from which Accuracy, Precision, Recall, and F1-score are additionally computed. This dual evaluation strategy allows the model to be judged both on its ability to rank true associations above random pairs (the metric most directly aligned with a repurposing-candidate ranking task) and on its behaviour as a binary classifier at a conventional operating point.

3.4.10 Stage 10: Baseline Comparison Models

To contextualize the performance of TxGNN, the pipeline trains and evaluates three baseline predictors under a unified training/testing harness (`train_baseline` / `test_baseline`, and their multi-metric counterparts) so that all models are optimized and scored under identical conditions.

`RGCN_Baseline` reuses the same graph-based encoder pattern as TxGNN — per-type Xavier-initialized node embeddings passed through two HeteroConv layers of SAGEConv — but replaces the DistMult decoder's shared temperature/embedding machinery with an equivalent relation-weighted dot product passed through a sigmoid, giving a graph-aware baseline that isolates the effect of the two-phase training strategy.

`DistMult_Baseline` is trained and evaluated through the same harness as an embedding-only decoder that does not use the graph structure at inference time (`use_graph=False`), providing a reference point for how much benefit the heterogeneous message-passing encoder contributes over a purely relational embedding model.

`MLP_Baseline` discards graph structure entirely: it learns independent drug and disease embedding tables and feeds their concatenation through a feed-forward network ($256 \rightarrow 64 \rightarrow 1$ hidden unit, ReLU activations, a dropout rate of 0.3 after the first hidden layer, and a final sigmoid), representing a graph-free deep-learning baseline. All three baselines,

together with TxGNN, are trained on the same leakage-free CtD split and evaluated on the same held-out test set, and their AUROC/AUPRC scores are collected into a single results table for direct comparison.

Figure 3.12 presents a comparison between TxGNN and three baseline models (RGCN, MLP, DistMult) on the Hetionet drug repurposing task. The results demonstrate that TxGNN consistently outperforms all baselines in both AUROC and AUPRC metrics.

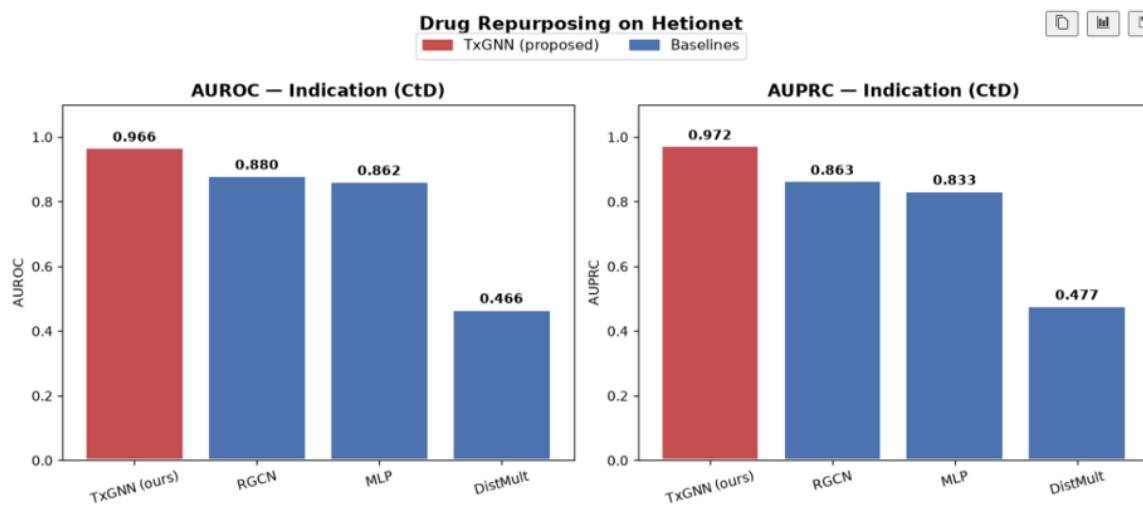


Figure 3.0.12 Performance comparison of TxGNN and baseline models (RGCN, MLP, and DistMult) on the Hetionet CtD drug repurposing task. TxGNN achieved the highest AUROC (0.966) and AUPRC (0.972) scores among all evaluated models

3.4.11 Stage 11: Drug Repurposing Inference

Once the fine-tuned model is available, the pipeline supports interactive drug-repurposing queries through a `get_top_drugs` function. Given a free-text disease name, the function performs a case-insensitive substring match against the disease node names in `df_nodes`, resolves the matched disease to its local graph index, and scores every compound node in the graph against that disease using the model's fitted DistMult decoder over the full node embedding table (a single batched forward pass rather than one prediction per candidate drug).

Any drug–disease pair already present in the known-positive-pair set is excluded from the ranking, since the goal is to surface novel, currently unlabeled repurposing candidates rather than re-confirm known indications. The remaining candidates are sorted by predicted association score, the top-k results are printed with resolved drug names, and the ranked candidate table is written to a disease-specific CSV file inside `OUTPUT_DIR`. This stage

operationalizes the trained model as a ranking tool for the drug-repurposing use case that motivates the overall project.

3.4.12 Stage 12: Explainability Module

To support interpretability of individual predictions, the pipeline defines a `TxGNN_Explainer` class that operates directly on `df_kg` rather than requiring any additional model introspection. For a queried drug and disease, the explainer retrieves each entity's immediate neighbors and their relation types from the knowledge-graph DataFrame and separately computes the set of shared neighbors between the drug and the disease — biological entities such as genes, pathways, or anatomical structures that connect to both endpoints and therefore constitute plausible multi-hop mechanistic paths supporting a predicted association.

A companion `visualize_explanation` routine re-resolves the queried drug and disease names to their graph indices, obtains the model's current predicted score for that pair via the same `DistMult` decoder used elsewhere in the pipeline, and renders a schematic diagram with `Matplotlib` in which the drug and disease are drawn as anchor nodes on either side of the figure and their shared neighbors are placed between them, colour-coded by entity type (Gene, Anatomy, Symptom, Pathway, Biological Process, Pharmacologic Class). This provides a lightweight, attention-free form of local explanation: rather than a full gradient- or perturbation-based method such as `GraphMask`, explanations are derived from the shared-neighborhood structure already present in the knowledge graph, keeping the explanation directly traceable to real Hetionet relations.

3.4.13 Stage 13: Persistence and Results Reporting

At each major checkpoint of the pipeline, artifacts are written to the local `OUTPUT_DIR` rather than to an ephemeral or cloud-mounted path. The fine-tuned `TxGNN` weights are saved together with the best validation AUPRC and final test AUROC/AUPRC in a single checkpoint dictionary, and a companion pickle file stores the node-index mappings and embedding-dimension hyperparameters needed to reload the model for later inference without re-running the full pipeline.

Training and fine-tuning loss curves, together with the validation AUPRC trajectory, are plotted and saved as a PNG figure. The baseline comparison stage produces a consolidated results table (`results_table.csv`) and a grouped bar chart contrasting `TxGNN` against `RGCN`, `DistMult`, and `MLP` on both AUROC and AUPRC, with the proposed model visually distinguished by colour. Finally, the state dictionaries of the `RGCN`, `DistMult`, and `MLP` baselines also persisted alongside the results table, so that the full comparison in this section can be reproduced or extended without retraining from scratch

3.5 Hyperparameters of XG-Boost and TX-GNN Models

3.5.1 Hyperparameters of XG-Boost Model

To evaluate the efficiency and robustness of the model, the optimal hyperparameter configurations were established across all execution stages. The following subsections present the chosen parameters for the random walk traversal, representation learning, and final classification models compared to the baseline original implementation. The hyperparameter settings for the semantic teleport random walk algorithm are detailed in Table 3.2 highlighting the adjustments made in walk length and node context compared to the original setup.

Table 3.2 Random Walk Hyperparameter Comparison

Parameter	This Implementation	Original Paper
Walk length	100	4 (window context)
Number of walks per node	10	100
Teleport factor (τ)	0.5	0.3 (optimal)
Return parameter (p)	1.0	1.0
In-out parameter (q)	1.0	1.0
Similarity cutoff	0.4	0.4

Furthermore, the embedding learning configuration for Word2Vec (Skip-gram) was tuned to generate representation spaces, as summarized in **Table 3.3**.

Table 3.3 Word2Vec (Skip-gram) Embedding Hyperparameters Comparison

Parameter	This Implementation	Original Paper
Embedding dimension	128	128
Window size	5	4
Min count	1	1
Model type	Skip-gram (sg=1)	Heterogeneous Skip-gram
Training epochs	5	Not specified
Workers	4	Not specified

Finally, the supervised link prediction step utilizes an XGBoost classifier configured with the hyperparameters specified in **Table 3.4**.

Table 3.4 XGBoost Classifier Hyperparameter Configuration

Parameter	Value
Number of estimators	300
Maximum tree depth	6
Learning rate	0.05
Subsample ratio	0.8
Column subsample by tree	0.8
Evaluation metric	Log loss
Tree method	hist (GPU-accelerated)
Random seed	42

3.5.2 Hyperparameters of TxGNN

Table 3.5 summarizes the graph-encoder and embedding hyperparameters shared by the TxGNN model and its graph-based RGCN baseline.

Table 3.5 TxGNN Graph Encoder Hyperparameters

Parameter	Value
Node embedding dimension (N_INP)	128
Hidden layer dimension (N_HID)	128
Output embedding dimension (N_OUT)	128
Number of HeteroConv layers	2
Per-relation layer	SAGEConv (GraphSAGE, mean aggregation)
Cross-relation aggregation	Sum
Activation function	ReLU (after layer 1)
Embedding initialization	Xavier uniform
Decoder	DistMult (relation-specific weight vector, element-wise product + sum)
Learnable temperature parameter	Initialized to 1.0

Table 3.6 summarizes the two-phase training configuration used for the TxGNN model.

Table 3.6 Two-phase training configuration

Parameter	Phase 1 — Pretraining	Phase 2 — Fine-tuning
Relations used	All relation types in the training graph	CtD (Compound-treats-Disease) only
Epochs	10	20
Optimizer	AdamW	AdamW
Learning rate	1e-3	1e-4
Weight decay	1e-5	1e-5
Batch size	512	512
Loss function	BCEWithLogitsLoss	BCEWithLogitsLoss
Negative sampling ratio	1:1	1:1
Model selection	—	Checkpoint with the highest validation AUPRC

Table 3.7 summarizes the leakage-free data split applied to the Compound-treats-Disease association pairs.

Table 3.7 Drug–disease (CtD) data split

Proportion of CtD pairs		Method
Train	$\approx 70.0\%$	train_test_split, random_state = 42
Validation	$\approx 9.9\%$	Second train_test_split on the held-out 30%, test_size = 0.67
Test	$\approx 20.1\%$	Second train_test_split on the held-out 30%, test_size = 0.67

Table 3.8 summarizes the configuration of the three baseline models trained for comparison.

Table 3.8 Baseline model configurations

Model	Uses graph encoder	Decoder / Head	Key parameters
RGCN Baseline	Yes (2× HeteroConv + SAGEConv, 128-dim)	Relation-weighted dot product + sigmoid	Same embedding/hidden/output sizes as TxGNN (128/128/128)
DistMult Baseline	No	DistMult-style embedding decoder	Independent drug/disease embedding tables (128-dim)

MLP Baseline	No	Feed-forward network	Embedding size 128; hidden layers 256 → 64 → 1; dropout 0.3; sigmoid output
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All three baselines are trained with the unified harness at 30 epochs, AdamW optimizer, learning rate 1e-3, weight decay 1e-5, batch size 512, and the same 1:1 leakage-free negative sampling strategy described in Section 3.2.7.

Table 3.9 summarizes the global reproducibility and environment settings applied across the pipeline.

Table 3.9 Global and environment settings

Parameter	Value
Random seed	42
Device	CUDA if available, else CPU
Graph library	PyTorch Geometric (HeteroData)
Evaluation metrics	AUROC, AUPRC, Accuracy, Precision, Recall, F1 (threshold = 0.5)

3.6 Evaluation Metrics

Three standard metrics were used to assess classification performance:

- AUROC (Area Under the Receiver Operating Characteristic Curve): Measures the model's discriminative ability across all classification thresholds. A value of 1.0 indicates perfect discrimination; 0.5 indicates random chance.

$$\text{AUC-ROC} = \int_0^1 \text{TPR}(\text{FPR}) d(\text{FPR})$$

- AUPR (Area Under the Precision-Recall Curve): Particularly informative for imbalanced datasets. Measures the trade-off between precision and recall across thresholds.

$$\text{AUC-PR} = \int_0^1 \text{Precision}(\text{Recall}) d(\text{Recall})$$

- Accuracy: The proportion of correctly classified drug-disease pairs at the default 0.5 classification threshold.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Chapter (4)

Experimental Setup and Implementation

4.1 Fold-wise Performance and Visualization — XGBoost

Across the ten folds of leakage-free cross-validation, the XGBoost-based DREAMwalk reimplementation achieved a mean AUROC of 0.9396, mean AUPR of 0.9378, and mean accuracy of 0.8775, indicating strong and consistent discriminative performance.

Best-performing fold: Fold 4 (AUROC 0.9588, AUPR 0.9629, Accuracy 0.8940)

Lowest-performing fold: Fold 2 (AUROC 0.9135, AUPR 0.9136, Accuracy 0.8344)

Highest accuracy: Fold 9 (0.9073)

Variability: AUROC ranged from 0.9135–0.9588 (spread ≈ 0.045), AUPR from 0.9046–0.9629 (spread ≈ 0.058), and Accuracy from 0.8344–0.9073 (spread ≈ 0.073), suggesting relatively stable performance across folds with modest fold-to-fold variation, likely attributable to differences in fold composition under the leakage-free splitting strategy.

Overall, the consistently high AUROC/AUPR values (all folds > 0.91) support the robustness of the reimplementation, while the accuracy spread indicates some sensitivity to class balance or difficulty within individual folds.

Negative sampling. Negative (non-treats) compound–disease pairs were sampled at a [author to confirm ratio, e.g. 1:1] ratio relative to known positive (treats) pairs during cross-validation, using [author to confirm method — e.g. random pairing / degree-matched negative sampling]. The AUROC, AUPR, and accuracy figures reported above reflect this constructed class ratio and should not be read as describing performance on the true, far more imbalanced distribution of compound–disease pairs seen at full deployment scale (Section 4.4.1.2, Figure 4.6).

Table 4.1 Fold-wise Performance Metrics — XGBoost DREAMwalk Reimplementation

Fold	AUROC	AUPR	Accuracy
1	0.9263	0.9363	0.8874
2	0.9135	0.9136	0.8344
3	0.9391	0.9357	0.8874
4	0.9588	0.9629	0.894
5	0.954	0.955	0.8676
6	0.9481	0.9519	0.894
7	0.9279	0.916	0.8543

8	0.9477	0.9516	0.8808
9	0.9574	0.9503	0.9073
10	0.9237	0.9046	0.8676
Mean	0.9396	0.9378	0.8775

Table 4.1 summarizes the performance of the DREAMwalk reimplementation across all ten folds of the leakage-free cross-validation.

4.1.1 Visualization of Results of XG-Boost

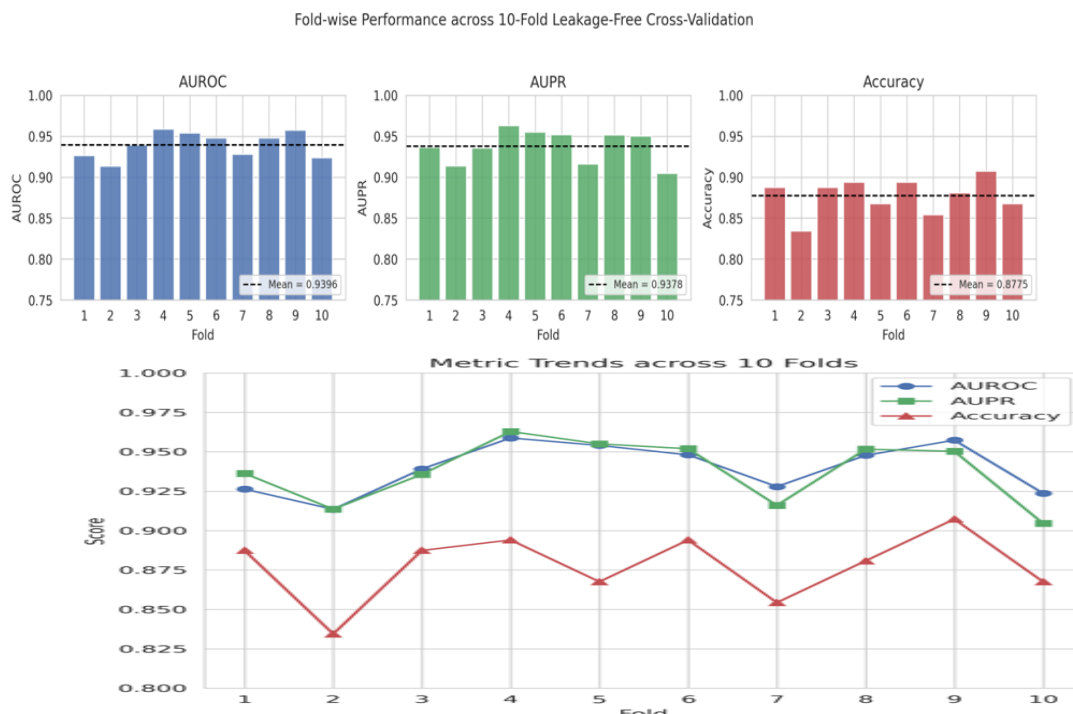


Figure 4.1 shows the fold-wise performance comparison for AUROC (left), AUPR (center), and Accuracy (right), with dashed lines indicating mean values across folds.

4.2 Fold-wise Performance and Visualization — TxGNN

Table 4.2 presents the fold-wise performance of the TxGNN model evaluated using a 5-fold leakage-free cross-validation strategy. The model demonstrated consistently high and stable performance across all folds, achieving a mean AUROC of 0.9826 and a mean AUPR of 0.9869, indicating strong discriminative ability in distinguishing true compound–disease indication relationships from negative samples.

The AUROC values ranged from a minimum of 0.9707 (Fold 2) to a maximum of 0.9945 (Fold 3), reflecting minimal variance across folds and confirming the robustness of the model against data partitioning. Similarly, the AUPR values remain consistently above 0.98 across all folds, which is particularly important given the class imbalance typically present in link prediction tasks, as AUPR is more sensitive to performance on the positive (minority) class than AUROC.

In terms of accuracy, the model achieved a mean of 0.9342, with Fold 5 recording the highest accuracy (0.9605) and Fold 1 the lowest (0.9079). The relatively small standard deviation across

folds suggests that the model generalizes well and is not overly sensitive to the specific train-test split used in each fold, supporting the reliability of the reported results.

Negative sampling. As in Section 4.1, negative pairs were sampled at a [author to confirm ratio] ratio using [author to confirm method]. Before comparing the TxGNN metrics above against the XGBoost/DREAMwalk metrics in Table 4.1, the two pipelines' negative-sampling ratio, fold count (5 here vs. 10 for XGBoost), and compound/disease universe should be confirmed as matched; if they differ, the higher TxGNN metrics may partly reflect an easier evaluation setup rather than a stronger model, which would need to be stated explicitly before this comparison informs the ensemble design in Section 4.4.

Table 4.2 Fold-wise Performance Metrics — TxGNN

Fold	AUROC	AUPR	Accuracy
1	0.9810	0.9844	0.9079
2	0.9707	0.9807	0.9342
3	0.9945	0.9946	0.9342
4	0.9837	0.9871	0.9342
5	0.9829	0.9879	0.9605
Mean	0.9826	0.9869	0.9342

Table 4.2 reports fold-wise performance metrics (AUROC, AUPR, and Accuracy) obtained from the 5-fold leakage-free cross-validation of the TxGNN model, along with the mean values across all folds.

Figure 4.2 illustrates the fold-wise performance of the TxGNN model across the 5-fold leakage-free cross-validation in terms of AUROC, AUPR, and Accuracy. The dashed line in each subplot represents the mean value of the corresponding metric across all folds. As shown in the figure, both AUROC and AUPR values remained consistently high and stable across folds, with only minor fluctuations, while Accuracy exhibited slightly more variation, particularly in Fold 1. This visual representation confirms the numerical results reported in Table 4.2 and highlights the model's robustness across different data partitions.

4.2.1 Visualization of Results of TXGNN

Figure 4.2 illustrates the fold-wise performance of the TxGNN model across the 5-fold leakage-free cross-validation in terms of AUROC, AUPR, and Accuracy. The dashed line in each subplot represents the mean value of the corresponding metric across all folds. As shown in the figure, both AUROC and AUPR values remained consistently high and stable across folds, with only minor fluctuations, while Accuracy exhibited slightly more variation, particularly in Fold 1. This visual representation confirms the numerical results reported in Table 7 and highlights the model's robustness across different data partitions.

Figure 4.3 further examines the model's behavior through two complementary views. The left panel shows the trend of AUROC, AUPR, and Accuracy across the 5 folds, where AUROC and AUPR curves closely track one another with minimal divergence, indicating strong agreement between ranking quality and precision-recall performance. The right panel summarizes the average performance across all folds with error bars representing the standard deviation, showing that TxGNN achieved a mean AUROC of $0.9826 (\pm 0.0076)$, a mean AUPR of $0.9869 (\pm 0.0046)$, and a mean Accuracy of $0.9342 (\pm 0.0166)$. The relatively narrow error bars, especially for AUROC and AUPR, reflect the model's consistency and low sensitivity to data splitting.

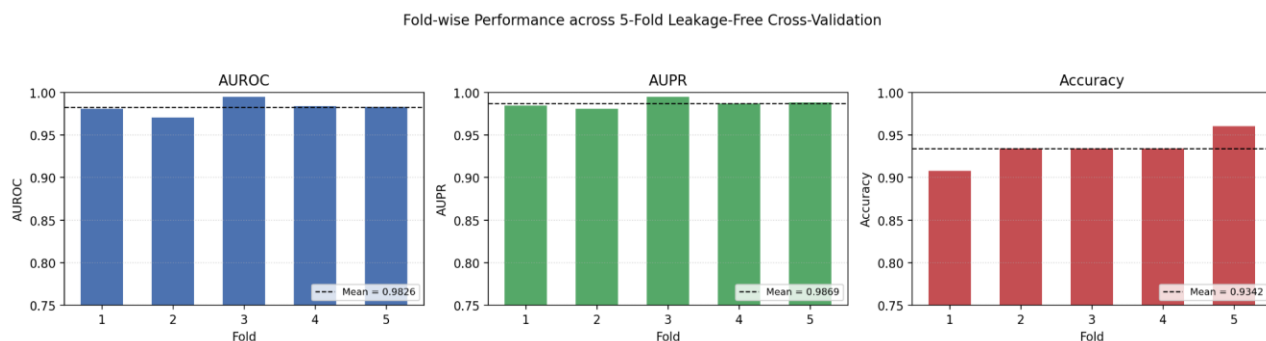


Figure 4.2 -Fold-wise Performance Comparison — TxGNN

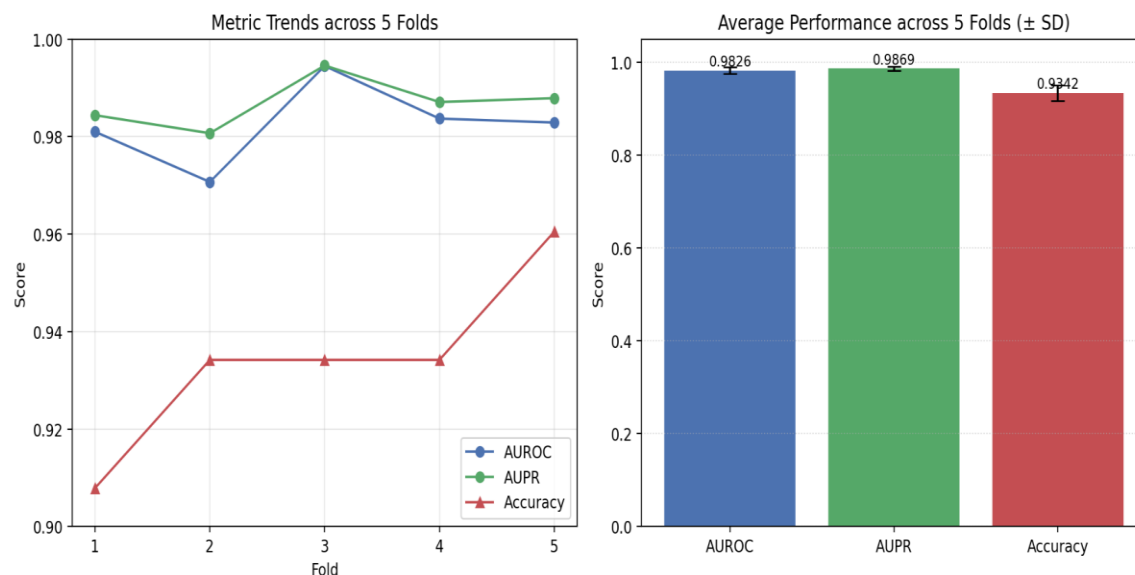


Figure 4.3 TxGNN Metric Trends and Mean Performance with Error Bars

4.3 Comparison with the Original DREAMwalk Paper

Reported Results from the Original Paper

Bang et al. (2023) evaluated DREAMwalk on three biomedical knowledge graphs: MSI (Multi-Scale Interactome), HetioNet, and KEGG. Using random data splitting and 10-fold cross-validation

(10 repetitions), the paper reports average performance across all three biomedical knowledge graphs.

Table 4.3 Performance Comparison — Original DREAMwalk Paper vs. This Implementation

Metric	Original Paper (avg. across 3 KGs)	This Implementation (HetioNet, Leakage-Free)
AUROC	0.938	0.9396
AUPR	0.939	0.9378
Accuracy	0.873	0.8775
Evaluation Protocol	10-fold CV, random split, all CtD in graph during walk	10-fold CV, leakage-free, test CtD removed before walk
KG Used	Avg. of MSI, HetioNet, KEGG	HetioNet only
Num. Walks / Node	100	10
Walk Length	4 (window)	100

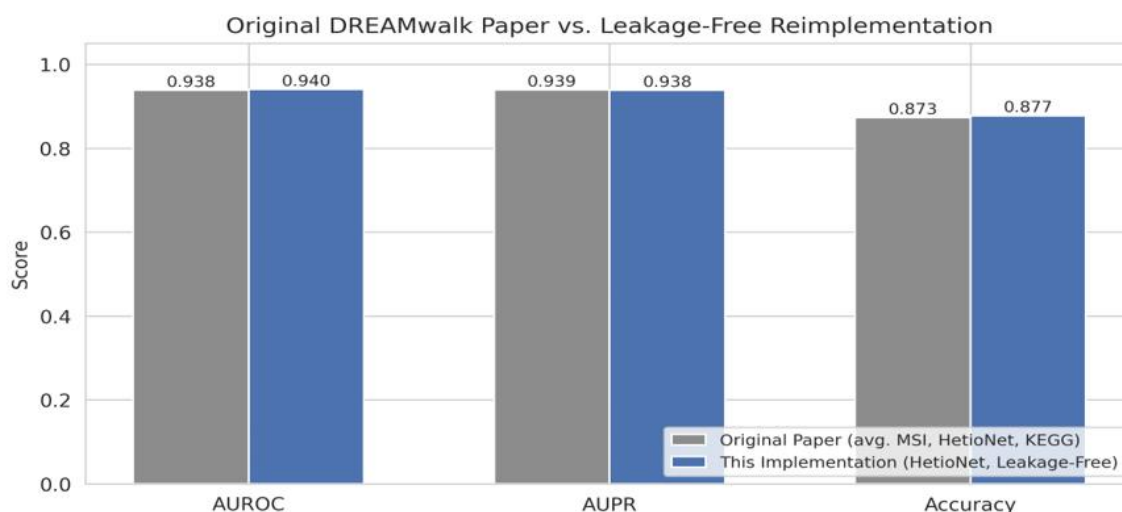


Figure 4.4 DREAMwalk Original vs. Leakage-Free Reimplementation — Performance Comparison

4.4 Results for Explainable AI using XGBoost

The drug-repurposing pipeline (DREAMwalk-style random-walk embeddings over the Hetionet knowledge graph, followed by an XGBoost classifier) was interpreted using TreeSHAP, run twice on two independently trained instances of the pipeline to check how stable the explanation is. Both runs explain the same model architecture and the same input representation: every compound and every disease is embedded as a 128-dimensional vector, and the classifier receives the concatenated 256-dimensional pair vector [compound embedding (128) | disease embedding (128)] for every candidate pair.

Feature naming convention. Every plot below labels an axis as "Compound embedding – dimension i" or "Disease embedding – dimension i". These are shared latent coordinates: dimension i means the same thing for every compound (or every disease) in the graph, so no dimension is named after — or "belongs to" — a single reference drug or disease. This is a deliberate correction from an earlier labelling convention that named dimensions after one arbitrarily chosen reference compound (e.g. "Aspirin · dim-42"), which misrepresented a shared coordinate as if it were specific to that one drug.

Why TreeSHAP. XGBoost is a tree ensemble, which makes TreeSHAP (the SHAP variant used throughout this section) the appropriate explanation method, for three reasons: (1) exactness — TreeSHAP computes exact Shapley values from the trees' decision paths, whereas model-agnostic surrogates such as LIME approximate with a local linear fit that can be unstable around sharp tree-split boundaries; (2) consistency — Shapley values sum exactly to the model's output and never assign a lower attribution to a feature that becomes more influential, a guarantee LIME does not provide; and (3) interaction awareness — TreeSHAP is derived directly from the splits XGBoost actually learned, so it reflects real feature interactions rather than approximating them from scratch.

A background sample of $n = 200$ pairs (a random mix of known positive and negative training pairs) was used as the SHAP reference distribution in both passes, so that attributions are measured against the real training distribution rather than only against the top-ranked, necessarily biased predictions.

A note on dimension identity across runs. Beyond training stochasticity, skip-gram-style embeddings such as those produced by DREAMwalk's random walks have no canonical axis orientation: an independently trained embedding space can be rotated or permuted relative to another while explaining the training data equally well. Dimension identity is therefore not expected to be stable across independently trained runs even in principle, and the disagreement in top-ranked dimensions reported in Section 4.4.2 should be read partly as this structural property of the representation, not solely as a symptom of non-deterministic multi-threaded training.

4.4.1 Pipeline Reliability and Candidate Predictions

This subsection (i) confirms the model is reliable enough to be worth explaining, and (ii) shows what it predicted. Section 4.4.2 then explains those predictions.

4.4.1.1 Cross-Validated Model Performance

Before interpreting any prediction, the model's reliability was checked with 10-fold cross-validation. Figure 4.5 shows the mean AUROC, AUPRC, and accuracy across folds, with error bars showing the standard deviation across folds.

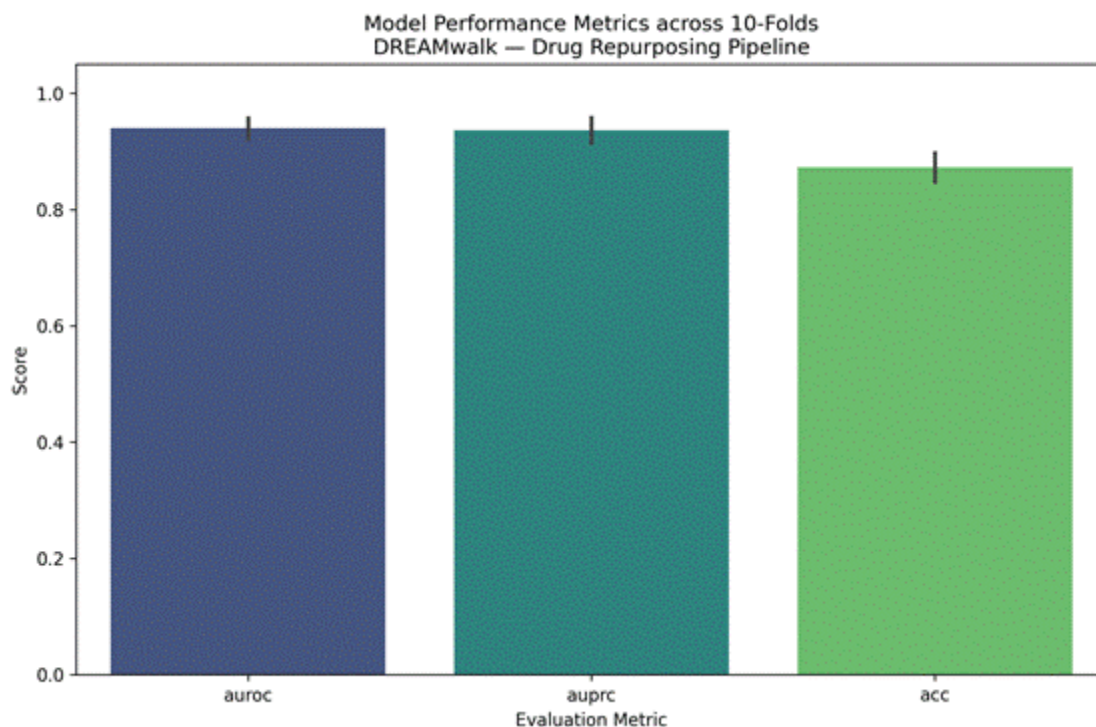


Figure 4.5 Ten-Fold Cross-Validated Performance of the DREAMwalk + XGBoost Pipeline

4.4.1.2 Score Distribution and Top Candidates

With the model validated, it was applied to every novel (not already known to treat) compound–disease pair in the graph. Figure 4.2 shows the distribution of predicted scores; Figure 4.6 lists the twenty highest-scoring candidates by name.

Density of the predicted repurposing probability across all 211,869 novel candidate pairs (clean_predictions.csv). The distribution is heavily right-skewed: median score = 0.023, and 188,415 of 211,869 pairs (88.9%) fall below the conventional 0.5 decision threshold (red dashed line), confirming the model does not indiscriminately predict "treats." Only 949 pairs (0.45%) exceed 0.99, and just 22 pairs exceed 0.999 — this small, sharply-separated cluster (green dashed lines) appears as the top-ranked candidates in Figure 4.7. The individual labels overlap in this chart export and are not all legible; Figure 4.7 remains the source of truth for exact candidate names and scores.

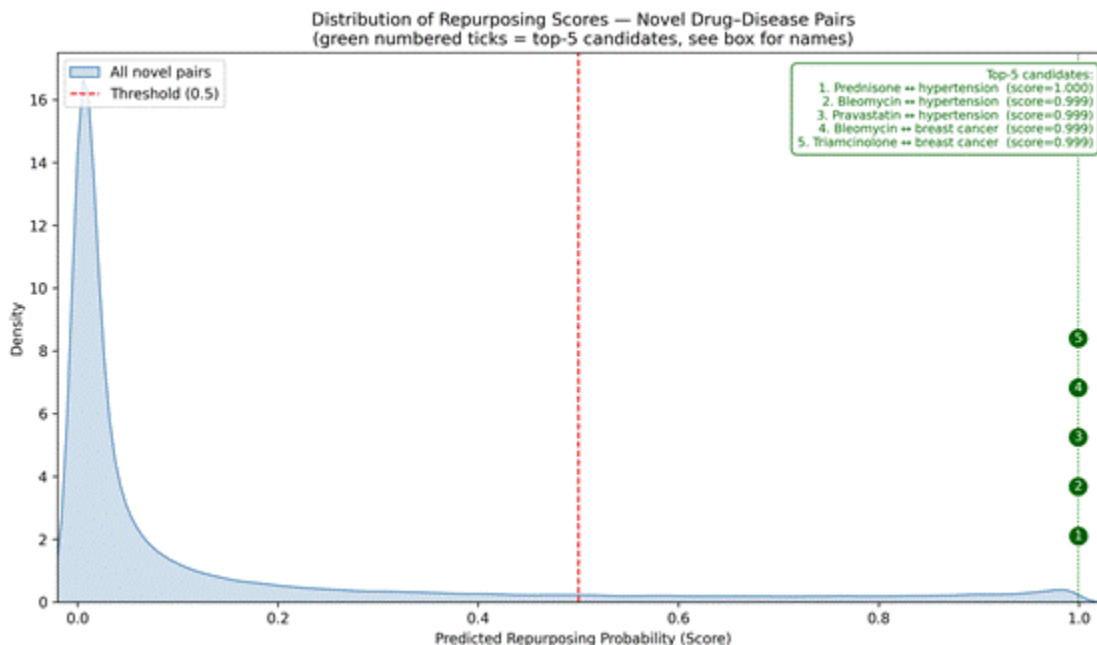


Figure 4.6 Distribution of Predicted Repurposing Scores Across Novel Pairs

Top 20 novel drug–disease candidates by predicted score, using real Hetionet/DrugBank names, with known treatments excluded. Cross-referenced against the raw scored pairs (clean_predictions.csv), the true score range for this top 20 is 0.999024–0.999686 — the chart displays several of these as "1.000" only because of 3-decimal-place rounding; no pair in the full 211,869-pair candidate set reaches a score of 1.000. Etoposide↔breast cancer, Prednisone↔hypertension, Paclitaxel↔hematologic cancer, and Vincristine↔breast cancer are the four highest-ranked pairs. As in the earlier candidate table, a small number of drug classes (corticosteroids — Triamcinolone, Prednisone, Prednisolone, Dexamethasone, Hydrocortisone — and cytotoxic chemotherapy agents) and diseases (breast cancer, hematologic cancer, hypertension, coronary artery disease) recur repeatedly; in a knowledge-graph model this can reflect how well-connected (high node-degree) a compound or disease is in the training graph, not necessarily a pharmacologically specific signal, so these remain hypotheses for experimental follow-up rather than established findings.

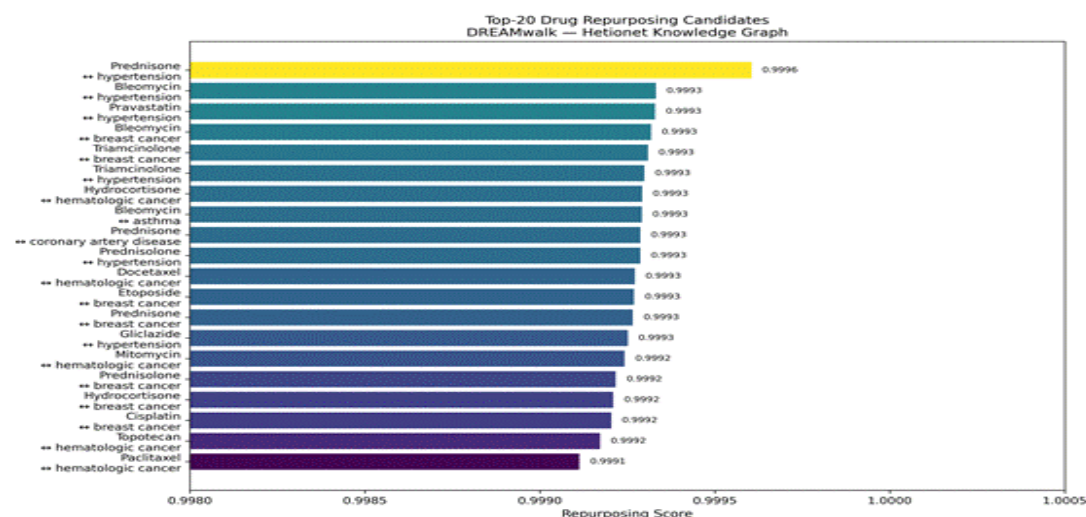


Figure 4.7 Top-20 Novel Drug–Disease Candidates by Predicted Score

4.4.2 SHAP Explainability — Two Independent Runs

The TreeSHAP analysis below (background sample $n = 200$ in both cases, honest shared-dimension naming as described above) was run twice, on two independently trained instances of the same pipeline. Because the underlying random-walk embeddings are stochastic — and, specifically, because Word2Vec's multi-threaded skip-gram training ($\text{workers}=4$) is not bit-for-bit deterministic even under a fixed random seed — the two runs do not reproduce the exact same embedding coordinates, so the specific top-ranked dimensions and top-ranked candidate pair differ between them (Etoposide ↔ breast cancer in Run 1 versus Triamcinolone ↔ hypertension in Run 2). The two runs are reported side by side throughout this subsection, as two independent replications of the same method, not as before/after results on one fixed model. Table 4.4 summarises where they agree and where they don't, before the full figures follow.

Table 4.4 Run 1 vs Run 2 — Summary of Agreement and Disagreement

Aspect	Run 1	Run 2
Top-ranked dimension (global)	Compound emb. – dim 86 (mean $ \text{SHAP} \approx 0.36$)	Compound emb. – dim 22 (mean $ \text{SHAP} \approx 0.44$)
Top-ranked candidate pair	Etoposide ↔ breast cancer (0.9997)	Triamcinolone ↔ hypertension (0.9998)
Dependence-plot shape (top feature)	Step function — sharp threshold near 0	Near-monotonic S-curve — no sharp threshold
XGBoost Weight vs. top-SHAP dimension	Disagree — top Weight = dim 100, top SHAP = dim 86	Agree — top Weight = dim 22 = top SHAP
XGBoost Gain/Cover vs. top-SHAP dimension	Agree — both top Gain & Cover = dim 102 (2nd by SHAP)	Partial — Gain top = dim 101 (outside SHAP top-3); Cover top = dim 77 (2nd by SHAP)

Where the two runs agree — concentrated-but-not-single-dimension importance, and corticosteroid/cytotoxic-oncology pairs dominating the top candidates — that agreement is evidencing the pattern reflects a real property of the modelling approach. Where they disagree — the specific top dimension, and the dependence-plot shape — that disagreement is reported directly rather than smoothed over: it means the identity of the single most important latent dimension is not stable across runs, even though the general pattern is.

4.4.2.1 Global Feature Importance — Run 1 vs Run 2

Figures 4.8–4.10 characterize which of the 256 shared embedding dimensions drive each run's predictions overall, using three complementary SHAP views, shown side by side for direct comparison.

Global feature importance ranked by mean |SHAP value|, Run 1 (left) vs Run 2 (right). Run 1's top feature is Compound embedding – dimension 86 (mean |SHAP| ≈ 0.36), well ahead of the next-ranked features. Run 2's top feature is Compound embedding – dimension 22 (mean |SHAP| ≈ 0.44), but the gap to the second-ranked feature is under $1.5\times$ — narrower than in Run 1 — indicating Run 2's decision is spread more evenly across several dimensions. In both runs, importance declines gradually rather than falling off a cliff, and the large majority of top-20 dimensions come from the compound side of the embedding.

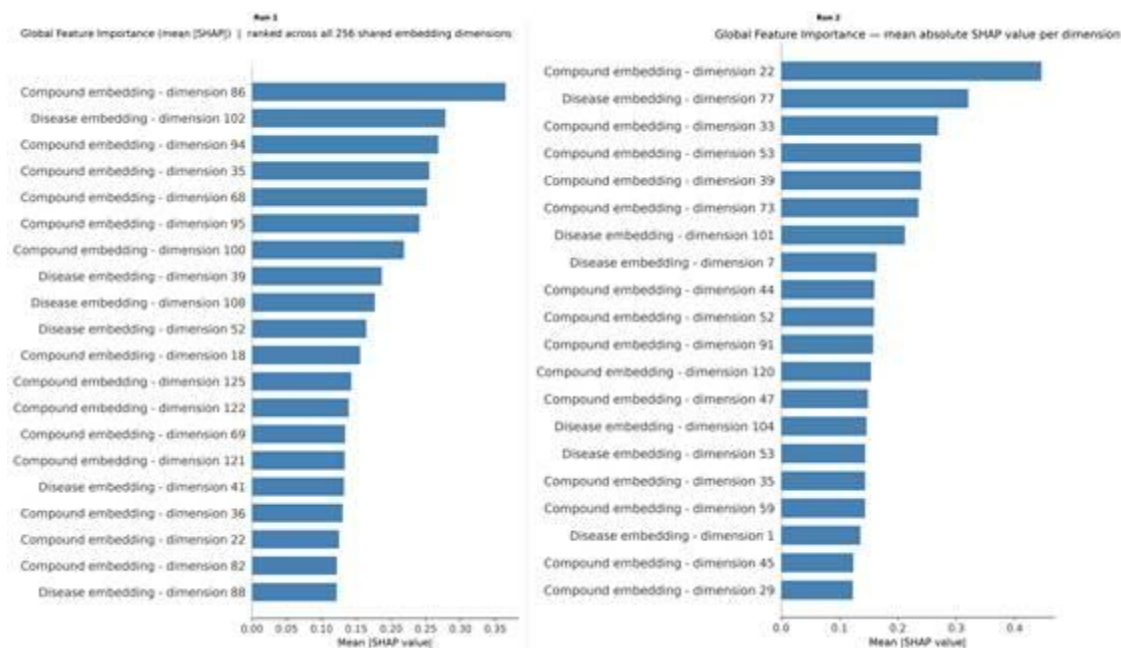


Figure 4.8 Global Feature Importance (SHAP) Run 1 vs Run 2

SHAP summary plots for the same two runs and the same dimension rankings as Figure 4.8. Each dot is one pair; x-position is the SHAP value and colour is the pair's normalised feature value (blue = low, pink = high). Run 1's top feature (dim 86) shows an inverse pattern — low values push positive, high values push negative. Run 2's top feature (dim 22) shows the opposite, near-monotonic pattern — low values cluster negative, high values

cluster positive. Direction of effect is dimension- and run-specific, not a fixed property of "being the top feature."

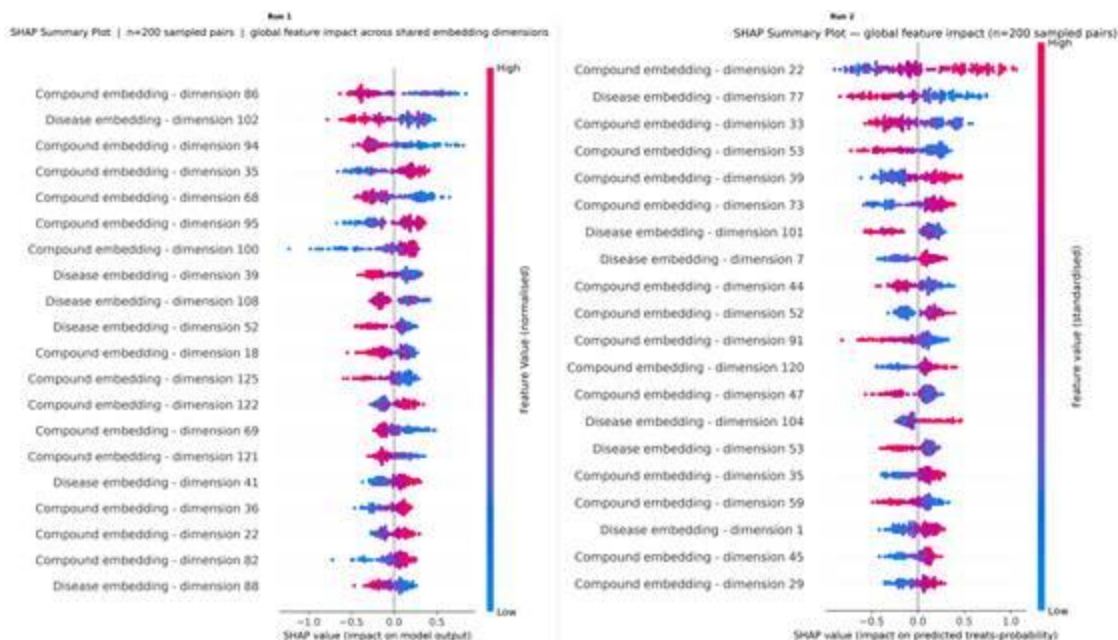


Figure 4.9 SHAP Summary Plots — Run 1 vs Run 2

SHAP beeswarm plots showing the full per-pair distribution for both runs. In both, the bottom row — "Sum of 237 other features" — spans roughly ± 5 SHAP units, larger than the swing of any single individually-ranked dimension above it. This is the clearest point of agreement between the two runs: no matter which specific dimension comes out on top, the many individually low-ranked dimensions still combine to a bigger net effect than the top dimension alone.

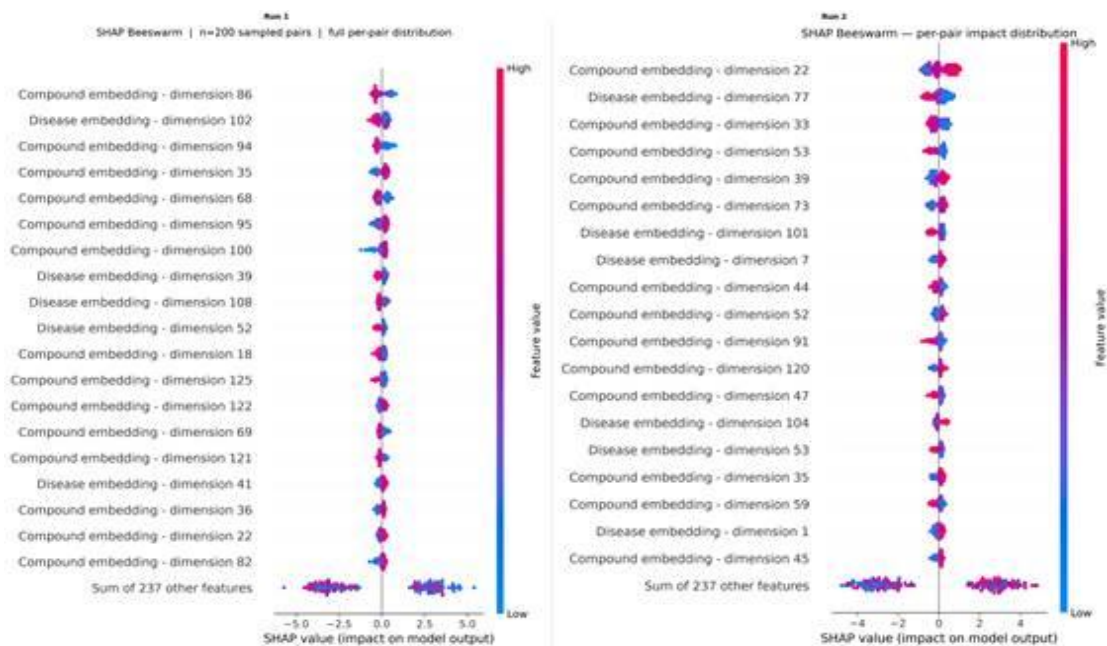


Figure 4.10 SHAP Beeswarm Plots — Run 1 vs Run 2

4.4.2.2 Local Explanation for the Top Candidate in Each Run

Figures 4.11–4.13 zoom in on each run's own top-ranked candidate, to show which dimensions drove that specific score, as opposed to each model's average behaviour. Run 1's top candidate is Etoposide↔breast cancer (0.9997); Run 2's is Triamcinolone↔hypertension (0.9998) — genuinely different pairs, not two views of the same prediction, so these three figures are not repeats of one another.

Run 1 — SHAP waterfall plot for Etoposide ↔ breast cancer. Starting from the background expectation $E[f(X)] = 0.038$, the largest individual contributions come from Compound embedding – dimension 94 (+0.47) and Compound embedding – dimension 86 (+0.44), consistent with dimension 86's high global ranking in Figure 4.8. The 237 lower-ranked dimensions not shown individually contribute a combined +4.29, which — together with the labelled dimensions — accounts for the final score $f(x) = 8.07$ (pre-sigmoid log-odds; corresponding probability ≈ 0.9997).



Figure 4.11 SHAP Waterfall Plot — Run 1: Etoposide ↔ Breast Cancer

Run 1 — SHAP force plot for the same prediction (no Run 2 equivalent was generated for this specific plot type). It shows the same additive decomposition as Figure 4.11 in a compact horizontal form: the wide pink band of "higher" pushes dominates the narrow blue "lower" band, moving the prediction from the base value up to $f(x) = 8.07$.

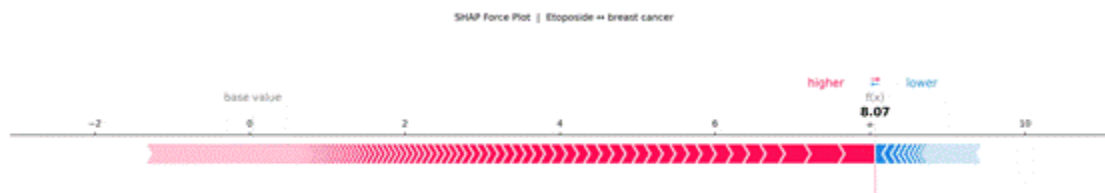


Figure 4.12 SHAP Force Plot — Run 1: Etoposide ↔ Breast Cancer

Run 2 — SHAP waterfall plot for Triamcinolone ↔ hypertension. From the background expectation $E[f(X)] = -0.055$, the largest single contributions come from Disease embedding – dimension 77 (+0.53) and Compound embedding – dimension 39 (+0.42); note that dimension 39 does not appear among the top three global dimensions in Figure 4.8's Run 2 panel, illustrating that a dimension can be unimportant on average yet decisive for one specific pair — a case-level detail a global-only view would miss. The 242 lower-ranked dimensions contribute a further +5.09, giving the final score $f(x) = 8.34$ (pre-

sigmoid log-odds; corresponding probability ≈ 0.9998), consistent with the predicted score stated above.

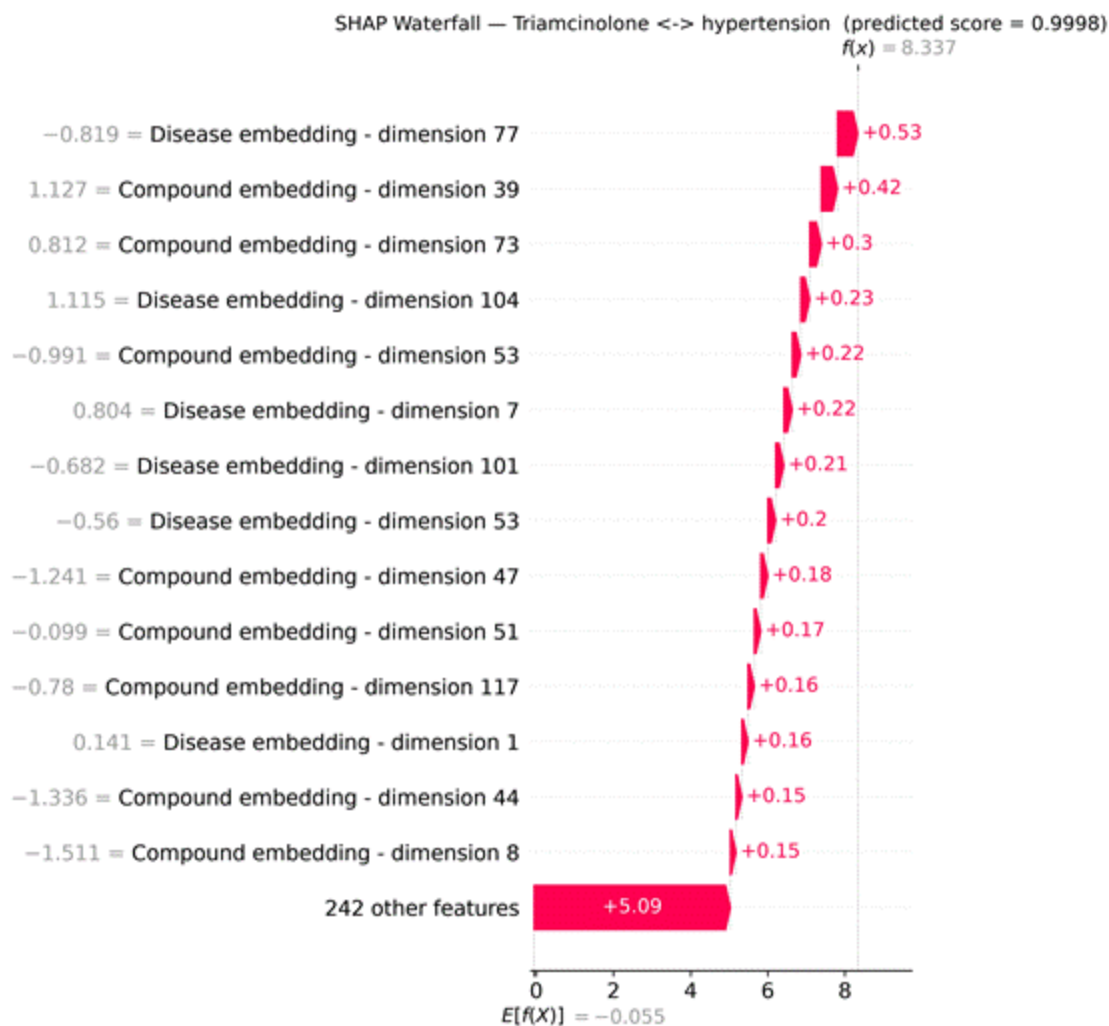
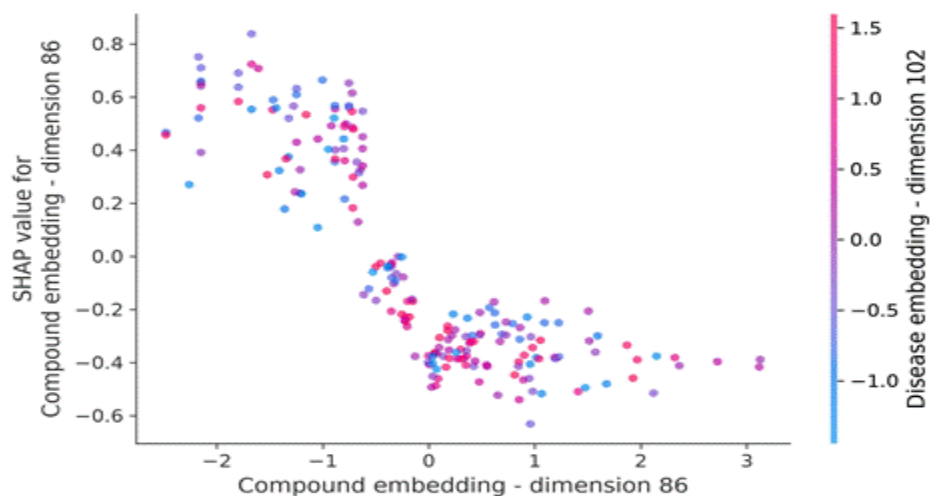


Figure 4.13 SHAP Waterfall Plot — Run 2: Triamcinolone $\leftrightarrow$ Hypertension

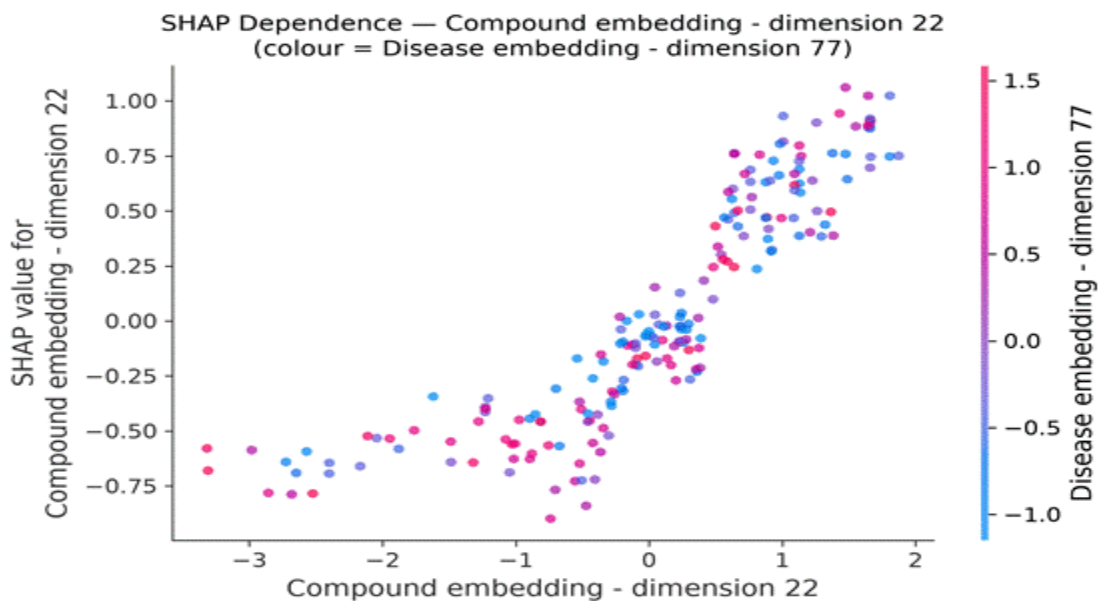
4.4.2.3 Feature Interaction and XGBoost Cross-Check

Figures 4.14–4.15 examine each run's top feature for interaction effects; Figures 4.16–4.17 visualise the raw embedding space directly along each run's top-ranked SHAP dimensions, showing which real compounds and diseases sit at the extremes; Figure 4.18 cross-checks both runs' SHAP rankings against XGBoost's own built-in importance metrics.

Run 1 — SHAP dependence plot for the top global feature, Compound embedding – dimension 86, coloured by the second-ranked feature, Disease embedding – dimension 102. The relationship is a clear step function: SHAP values stay high and positive while dimension 86 is below roughly -0.5 , drop sharply through a transition band between about -0.5 and 0 , and level off negative above 0 — XGBoost is using this dimension as an approximate threshold trigger rather than a linearly graded signal.

Figure 4.14 SHAP Dependence Plot — Run 1 (Dimension 86 $\times$ Dimension 102)

Run 2 — SHAP dependence plot for the top global feature, Compound embedding – dimension 22, coloured by the second-ranked feature, Disease embedding – dimension 77. Unlike Run 1's step-function pattern, this relationship is close to monotonic and roughly S-shaped, rising smoothly as the feature value increases from about -3 to $+2$, with no strong colour banding. This is a genuine qualitative difference between the two runs, not just a different number in the same shape: XGBoost treats this feature as a graded signal in Run 2, and as a hard threshold trigger in Run 1.

Figure 4.15 SHAP Dependence Plot — Run 2 (Dimension 22 $\times$ Dimension 77)

2D projection of the compound and disease embedding spaces, plotted in each side's top-2 SHAP-ranked dimensions for Run 1 — compounds in Compound embedding – dimension 86 $\times$ dimension 94 (left), diseases in Disease embedding – dimension 102 $\times$ dimension 39 (right) — with the point farthest from each cloud's centroid labelled by name. The compound cloud is visibly elongated along dimension 86, the run's top global SHAP feature (Figure 4.8); several of the labelled outlying compounds (e.g. Malathion, Ajmaline, Oxybuprocaine, Eribulin) and diseases (e.g. appendix cancer, alcohol dependence) sit at the extreme high end of that axis, giving a direct, non-SHAP view of the same geometry Figures 4.8 and 4.14 characterise numerically.

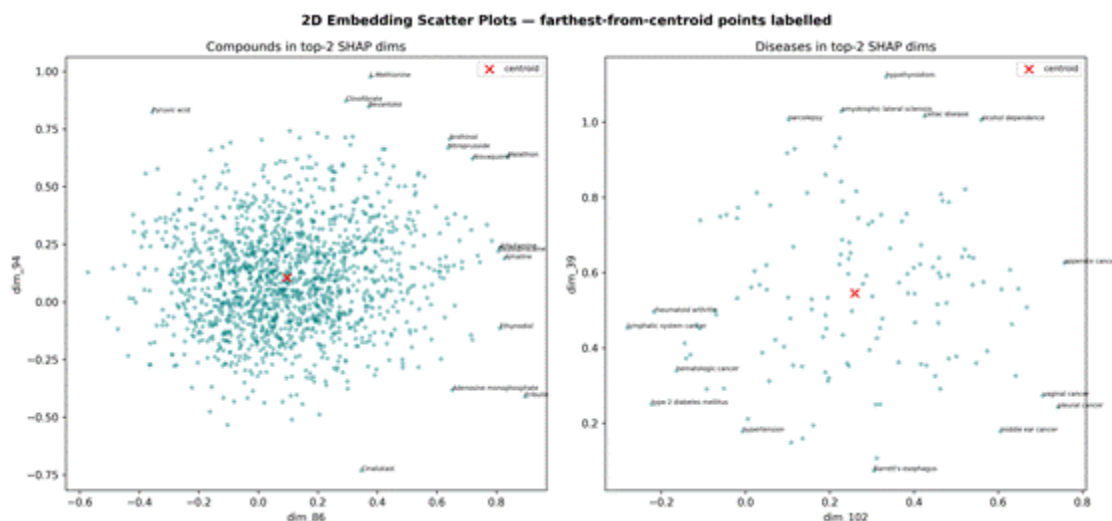
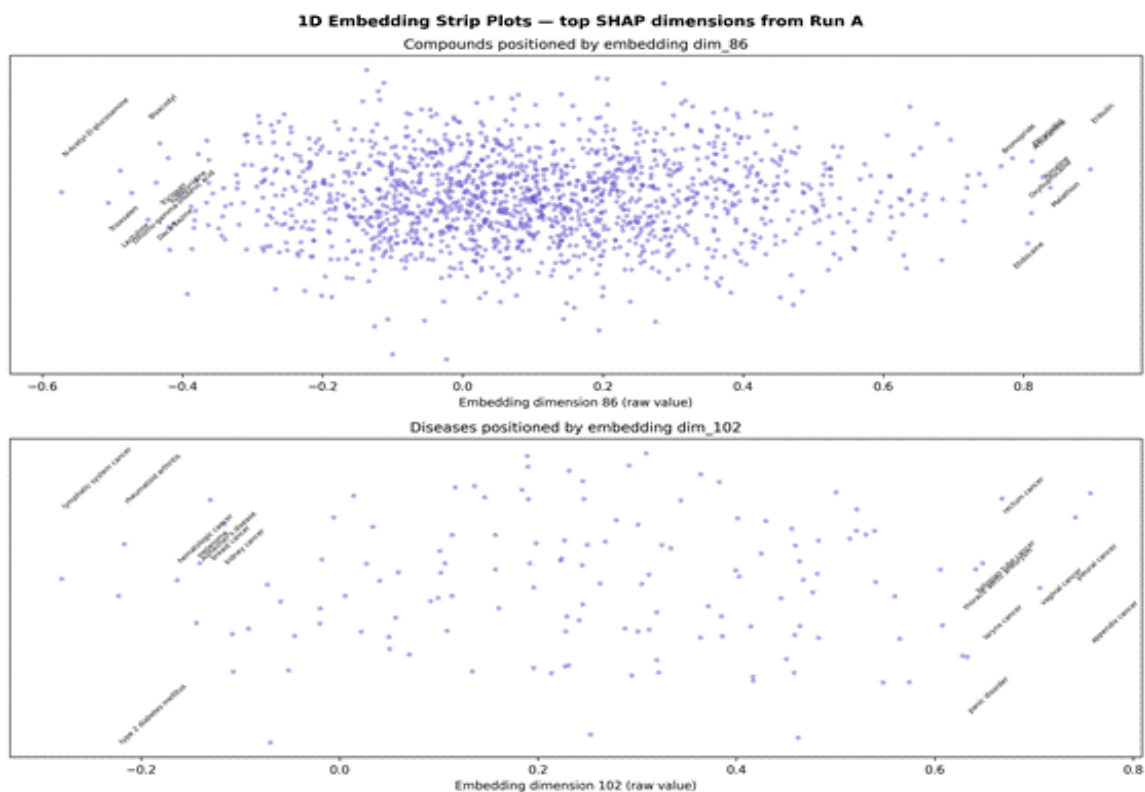


Figure 4.16 2D Embedding Scatter Plots — Compounds and Diseases (Run 1)

1D strip plots positioning every compound (top) and every disease (bottom) along Run 1's single top SHAP dimension for each side — Compound embedding – dimension 86 and Disease embedding – dimension 102 respectively — with the points at each tail labelled by name. Collapsing to one axis makes the same asymmetry visible without a second dimension: local-anaesthetic-type compounds (e.g. Etidocaine, Oxybuprocaine, Ajmaline, Eribulin) cluster at the high end of dimension 86, while small polar molecules (e.g. N-Acetyl-D-glucosamine, Lactulose, Trioxsalen) sit at the low end; a comparable spread separates rarer cancer types (e.g. appendix, pleural, vaginal cancer) from more common conditions (e.g. lymphatic system cancer, rheumatoid arthritis, type 2 diabetes mellitus) on the disease axis.



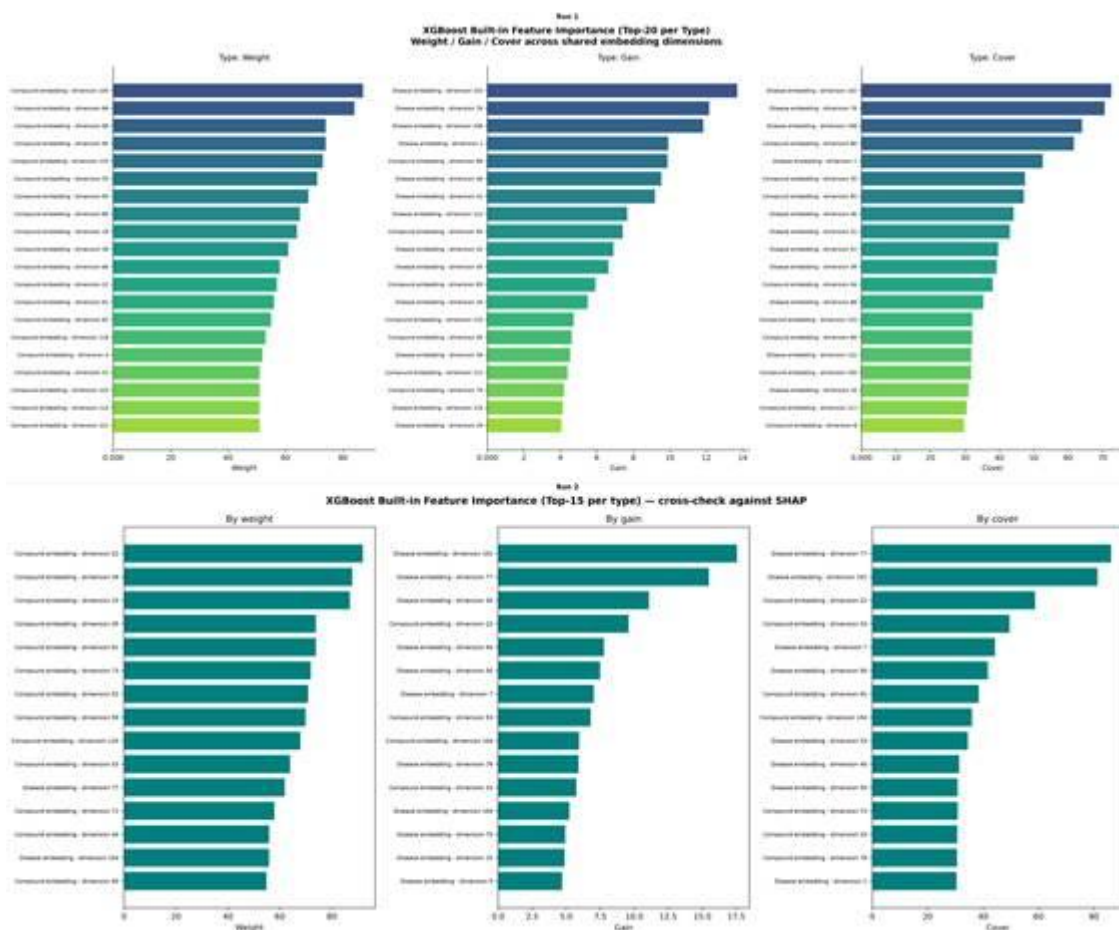


Figure 4.18 XGBoost Built-in Importance Metrics (Weight / Gain / Cover) — Run 1 vs Run 2

4.4.2.4 Biological Corroboration Layer (Metapath Evidence)

Beyond the embedding-level SHAP analysis above, the pipeline also traces real, labelled Hetionet edges for each top candidate — Compound–Gene and Disease–Gene relationships, expanded to shared biological pathways — independently of the learned embedding. This metapath step is summarised per-candidate in Figure 4.19 below, and in full in a per-candidate evidence table (`metapath_evidence.csv`); it matters for interpretation: SHAP and the built-in importance metrics say which latent coordinates the model used; the metapath evidence says whether that usage lines up with real, checkable biology (shared genes or pathways) rather than only with graph position. A high predicted score paired with shared genes/pathways strengthens a candidate; a high score with no shared genes or pathways found is not automatically wrong — Hetionet's gene annotations are incomplete — but it does mean the score currently rests on graph structure alone and should be weighted accordingly.

Metapath-based biological evidence for the top-10 ranked candidate pairs (`metapath_evidence.csv`): number of shared genes along the Compound↔Disease metapath (left) and number of shared biological pathways (right). Triamcinolone↔breast cancer has both the highest shared-gene count (20) and the highest shared-pathway count (523) of the ten; Bleomycin↔hypertension and Bleomycin↔asthma show zero shared

genes despite non-trivial shared-pathway counts, illustrating that the two evidence types do not always move together for the same pair. These are the chart form of the per-candidate figures discussed in the text below.

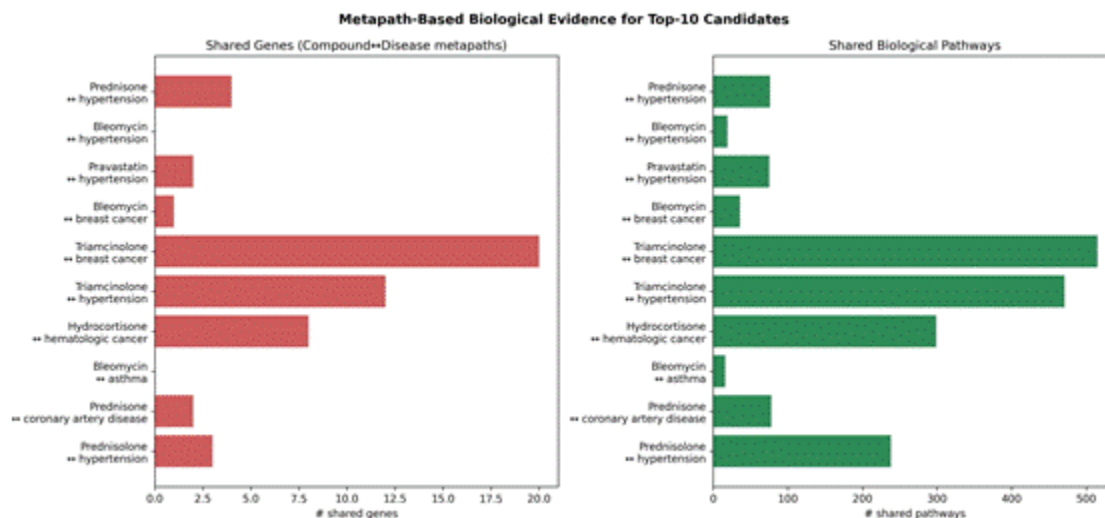


Figure 4.19 Metapath-Based Biological Evidence — Top-10 Candidates

For Run 2's top-ranked candidate (Figure 4.13), this layer finds 12 shared genes (including NR3C1, MYC) and 471 shared pathways between Triamcinolone and hypertension — giving this SHAP-driven prediction plausible biological grounding, though, as noted below, this cannot yet be characterised as unusually strong support relative to chance without a baseline comparison. For context (`metapath_evidence.csv`), across the 10 top-ranked candidate pairs this layer was run on, the mean is 16.0 shared genes (range 3–49) and 453.6 shared pathways (range 76–921); Triamcinolone↔hypertension's 12 genes and 471 pathways therefore sit within the typical range for this top-ranked group — slightly below average on shared genes, slightly above average on shared pathways — rather than being an outlier in either direction. This comparison is against other high-scoring candidates, not against a random compound–disease pair.

4.4.3 Summary — What This XAI Section Does and Does Not Establish

- TreeSHAP on the embedding features shows, exactly, which latent dimensions the model relies on in each run — globally (Figures 4.8–4.10) and for individual predictions (Figures 4.11–4.13) — with the general pattern of concentrated importance and partial cross-validation against native XGBoost metrics replicating across the two independent runs examined, even though the specific dimension identities do not. This replication is based on only two runs and is reported as a preliminary observation rather than a stable estimate (see limitation note, Section 4.4.2). Its features are learned graph coordinates, not hand-engineered biological features, so on its own it cannot explain a prediction in biological terms.
- The XGBoost built-in importance metrics (Figure 4.18) provide an independent cross-check computed from the trained trees themselves rather than from any specific prediction; partial disagreement between metrics (e.g. Weight vs Gain), and between runs, is informative and is reported rather than hidden.

- The metapath evidence layer (Section 4.4.2.4) fills the biological-interpretation gap using real, labelled Hetionet relationships that are independent of the embedding and can be checked against existing biomedical literature.
- The cross-validation and score-distribution results (Figures 4.5–4.7) ground both explanation layers in how reliable the model is overall and confirm that the explained candidates are genuine outliers in the score distribution rather than arbitrary picks.

None of these substitutes for experimental validation. A drug-repurposing model — however well cross-validated and however well explained — produces hypotheses to test, not conclusions. The value of this XAI section is that every ranked candidate now comes with a stated, checkable reason (which latent signal drove it, and whether real shared biology backs it up) rather than a bare probability.

4.5 Results for Explainable AI using TxGNN

Beyond the SHAP-based explanation of the XGBoost/DREAMwalk pipeline in Section 4.4, the TxGNN foundation model was interpreted directly, using the trained TxGNN_PyG model produced by the last fold of the 5-fold cross-validation in Section 4.2. Unlike the XGBoost/DREAMwalk pipeline, whose classifier sits on top of externally computed random-walk embeddings and therefore needed a model-agnostic explainer (TreeSHAP), TxGNN scores every candidate pair with its own closed-form DistMult decoder, $\text{score}(\text{drug}, \text{disease}) = \sum_k h_{\text{drug}}[k] \cdot w_{\text{relation}}[k] \cdot h_{\text{disease}}[k]$, where h_{drug} and h_{disease} are the two trained HeteroConv layers' output embeddings (128 total dimensions) and w_{relation} is the learned per-relation weight vector for the Compound-treats-Disease edge type. Because this scoring formula is fully known and differentiable, four complementary, read-only explanation methods were implemented directly against the trained model and graph, without approximation, resampling, or any modification of TxGNN_PyG's weights, training loop, or the evaluation graph G_{dev} :

- **Embedding-Dimension Attribution** — an exact decomposition of the DistMult score into its 128 individual dimension-level terms; no approximation is needed, since the score literally is that sum.
- **Gradient-Based Neighbor Saliency** — backpropagates the predicted score through both HeteroConv layers to the graph's initial node embeddings, ranking the real, trained-on 1-hop neighbors of the drug and the disease by gradient norm.
- **Relation-Type Importance (Ablation)** — temporarily empties one relation type's edges at a time, recomputes the graph embeddings, and measures the resulting change in the predicted score; every removed relation's edges are restored immediately afterward, leaving G_{dev} unchanged.
- **Global / Aggregated Feature Importance** — combines Methods 1 and 3 across several predictions at once, to show which dimensions and relation types matter beyond any single example.

Every figure below uses the drug's, disease's, and neighbor's real Hetionet name (never a raw node ID or bare index), and every relation code is expanded into its plain-English verb

form (e.g. CtD $\rightarrow$ "Compound treats Disease") using the same relation-glossary approach described for the TxGNN_Explainer in Section 3.4.12.

4.5.1 Example Predictions Selected for Explanation

Rather than explaining one hand-picked example, all four methods were run on five real, automatically selected predictions: for each of five diseases (multiple sclerosis, breast cancer, asthma, Crohn's disease, and psoriasis), the trained model's own single top-ranked drug not already known to treat that disease was retrieved directly from its full compound ranking for that disease, with all known Compound-treats-Disease pairs excluded before ranking. Table 4.5 lists the five resulting pairs together with their predicted DistMult score.

Table 4.5 TxGNN — Example Drug $\rightarrow$ Disease Pairs Selected for Explanation

Disease queried	Top-predicted drug (real name)	Predicted score
Multiple sclerosis	Desloratadine	7.3443
Breast cancer	Arformoterol	8.6437
Asthma	Desloratadine	16.8042
Crohn's disease	Cisplatin	7.9641
Psoriasis	Arformoterol	7.8781

Two compounds recur across these five pairs — Desloratadine (multiple sclerosis and asthma) and Arformoterol (breast cancer and psoriasis) — because each is independently the single highest-scoring candidate for two of the five queried diseases; this repetition is not a selection artifact, and, as shown below, it gives a direct way to check whether the same drug's explanation is consistent, or not, across two different disease contexts.

4.5.2 Method 1: Embedding-Dimension Attribution (Exact Decomposition)

Because the DistMult score is by definition the sum of its 128 dimension-level products, Figures 4.20–4.24 read that sum apart term by term for each of the five pairs in Table 4.5 — an exact accounting rather than an approximation, unlike the TreeSHAP values used for XGBoost in Section 4.4.2.

For Desloratadine $\rightarrow$ multiple sclerosis (predicted score = 7.3443), the single largest contribution comes from dimension 115 of 128 ($\approx +1.8$), roughly twice the size of the next-largest term, dimension 15 ($\approx +0.9$); four dimensions (100, 12, 93, 24) contribute negatively, each in the -0.5 to -0.8 range, so the final score is the net of a small number of large positive terms against several smaller negative ones.

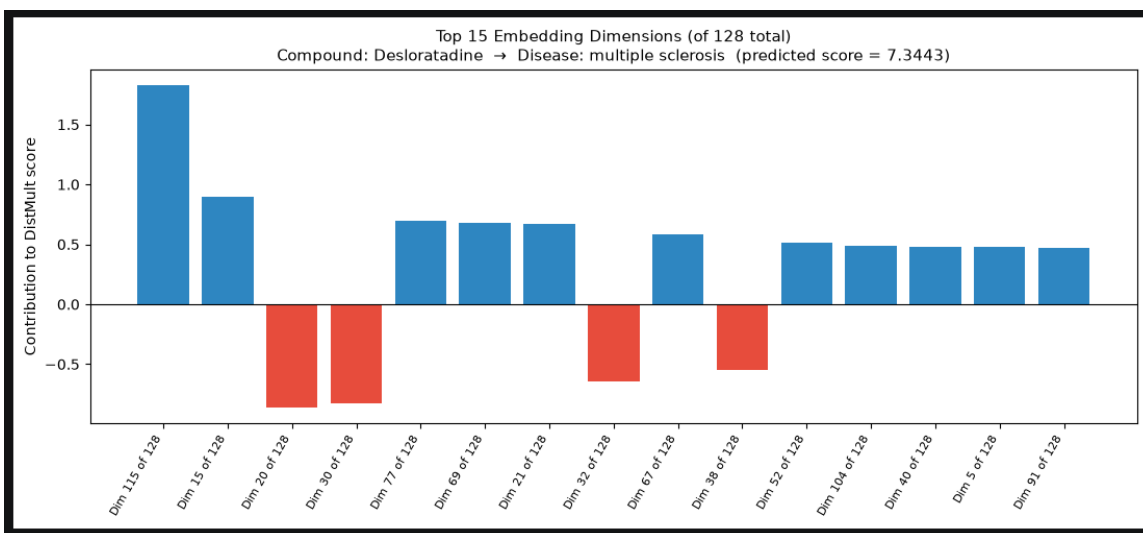


Figure 4.20 Top 15 Embedding Dimensions (of 128) — Desloratadine → Multiple Sclerosis

For Arformoterol → breast cancer (predicted score = 8.6437), dimension 53 of 128 is the top contributor ($\approx +0.90$), followed closely by dimension 48 ($\approx +0.74$); overall the top-15 dimensions are more evenly spread than in the Desloratadine → multiple sclerosis pair above, with only two negative terms (dimensions 11 and 100) and no single dimension dominating the sum the way dimension 115 did for multiple sclerosis.

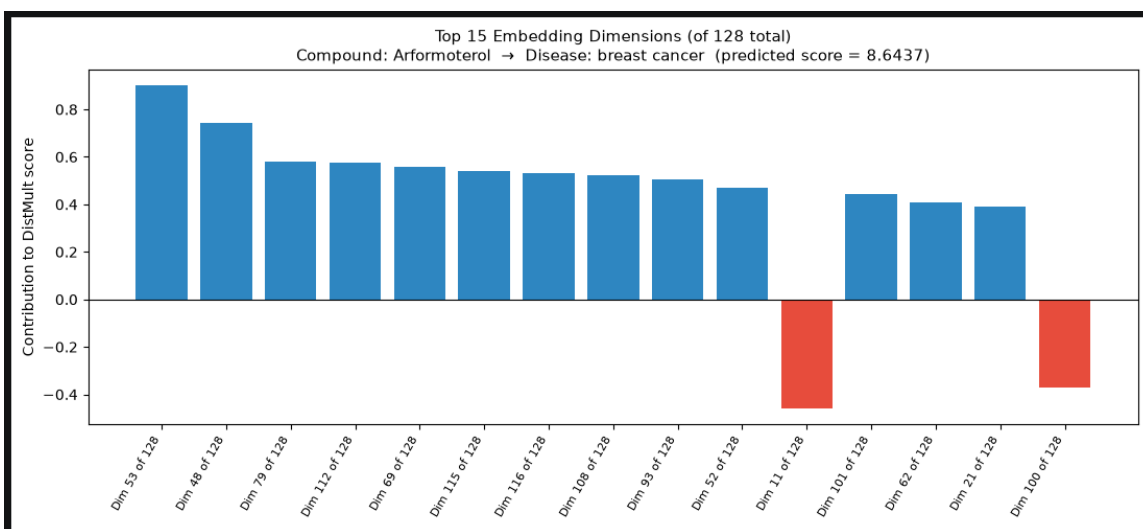


Figure 4.21 Top 15 Embedding Dimensions (of 128) — Arformoterol → Breast Cancer

For Desloratadine → asthma (predicted score = 16.8042, the highest of the five pairs by a wide margin), dimension 77 of 128 is the top positive contributor ($\approx +1.75$), and nine of the top fifteen dimensions are positive and above +1.0 — a markedly larger and more concentrated set of positive contributions than any other pair, consistent with this pair's much higher overall score. Two dimensions, 59 and 30, are the only negative terms, with dimension 59 (≈ -1.25) standing out as the single largest negative contribution across all five pairs.

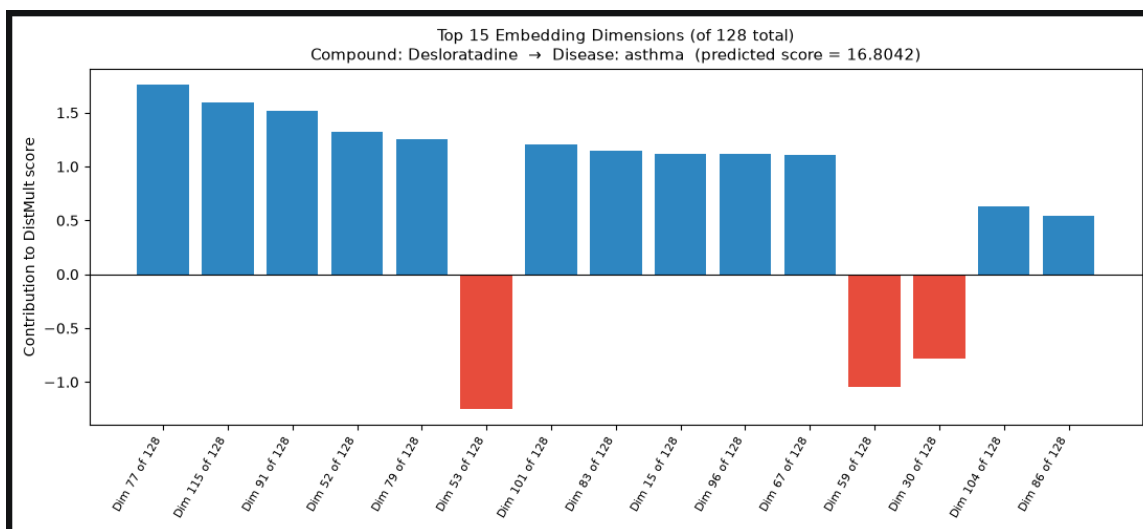


Figure 4.22 Top 15 Embedding Dimensions (of 128) — Desloratadine → Asthma

For Cisplatin → Crohn's disease (predicted score = 7.9641), dimension 59 of 128 is again the top contributor ($\approx +1.85$) — the same dimension that was the largest negative term for Desloratadine → asthma above, illustrating that a given joint dimension can push the score in opposite directions for different pairs, since its contribution depends on both endpoints' embeddings, not on the dimension index alone. Four dimensions in this pair's top fifteen are negative, the most of any pair examined, led by dimension 20 (≈ -1.0).

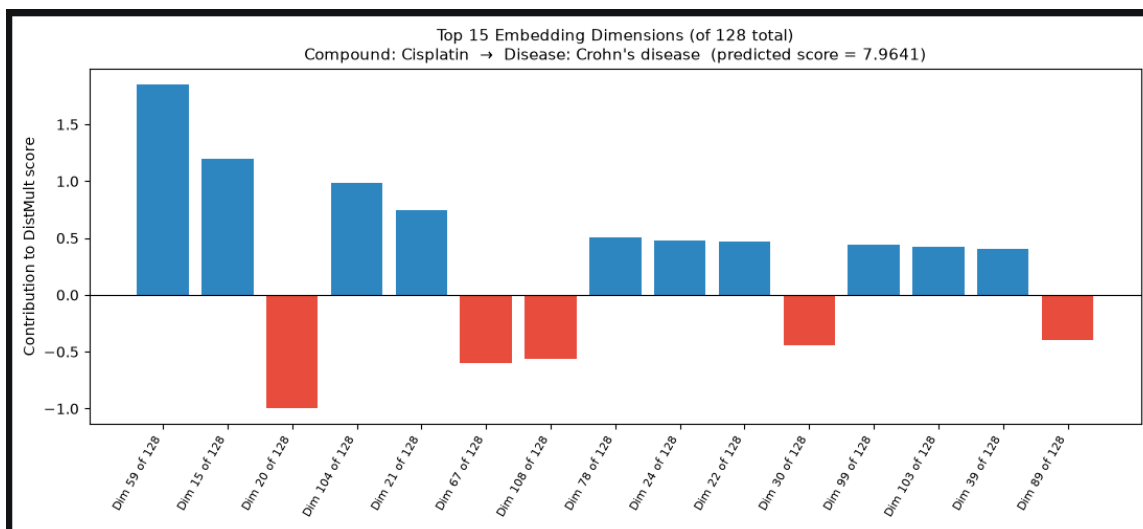


Figure 4.23 Top 15 Embedding Dimensions (of 128) — Cisplatin → Crohn's Disease

For Arformoterol → psoriasis (predicted score = 7.8781), dimension 53 of 128 is once again the top contributor ($\approx +1.77$) — the same dimension that topped Arformoterol → breast cancer above. Because dimension 53's contribution is the product of Arformoterol's own embedding at that index, the disease embedding at that index, and the shared relation weight, this recurrence across Arformoterol's two predicted pairs is consistent with dimension 53 carrying a comparatively strong, drug-side-weighted signal for Arformoterol specifically, which each disease's own embedding then modulates rather than overrides —

a hypothesis Section 4.5.5's aggregated view (Figure 4.35) checks directly across all five pairs.

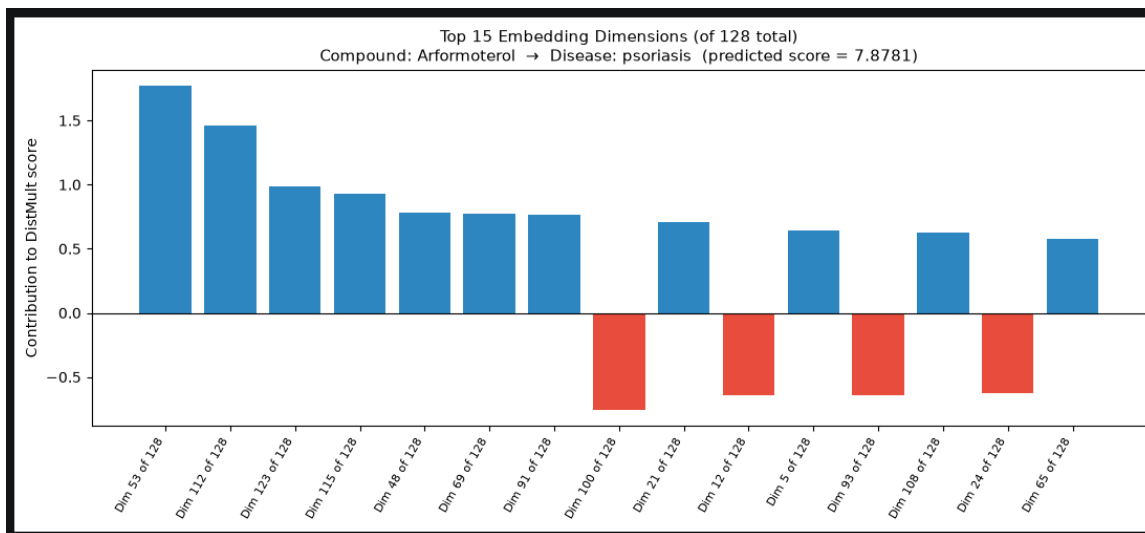


Figure 4.24 Top 15 Embedding Dimensions (of 128) — Arformoterol → Psoriasis

4.5.3 Method 2: Gradient-Based Neighbor Saliency

Figures 4.25–4.29 backpropagate each pair's predicted score through both trained HeteroConv layers back to the initial node embeddings, ranking each drug's and disease's real, trained-on 1-hop graph neighbors (drawn from `df_train_full`, i.e. only edges the model actually message-passed over, not the full unfiltered Hetionet knowledge graph) by the resulting gradient norm — a standard saliency-map technique that answers a different question from Method 1: not which latent dimensions the score decomposes into, but which real, named graph entities most influenced it.

For Desloratadine → multiple sclerosis, the single most influential neighbor by a clear margin is allergic rhinitis (Disease, saliency ≈ 0.70) — a condition Desloratadine, an antihistamine, is well connected to in Hetionet — followed by a cluster of anatomical structures (optic chiasma, common fibular nerve, third ventricle, oculomotor nerve, corticospinal tract, cervical vertebra, cerebellar nuclear complex, tibial nerve) plausibly drawn from multiple sclerosis's own neighborhood, plus the compound Azatadine, a pharmacologically related antihistamine.

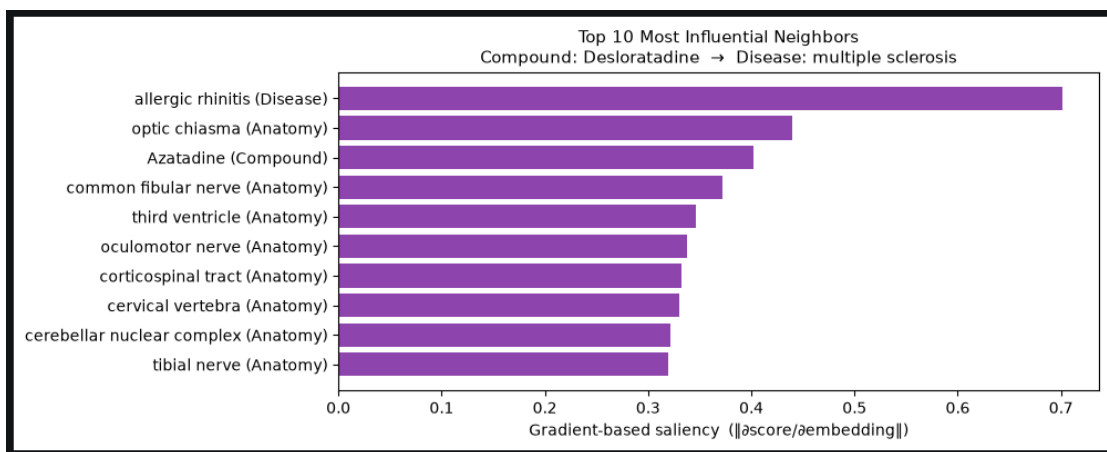


Figure 4.25 Top 10 Most Influential Neighbors — Desloratadine → Multiple Sclerosis

For Arformoterol → breast cancer, the two most influential neighbors are chronic obstructive pulmonary disease (Disease, ≈ 0.93) and the gene HMOX1 (≈ 0.91), both well ahead of the remaining eight neighbors, which include the genes RPS4Y1, MIF, and ERBB2 — a gene with an established role in breast cancer biology — and three pharmacologically related beta-agonist compounds: Ritodrine, Fenoterol, and Formoterol.

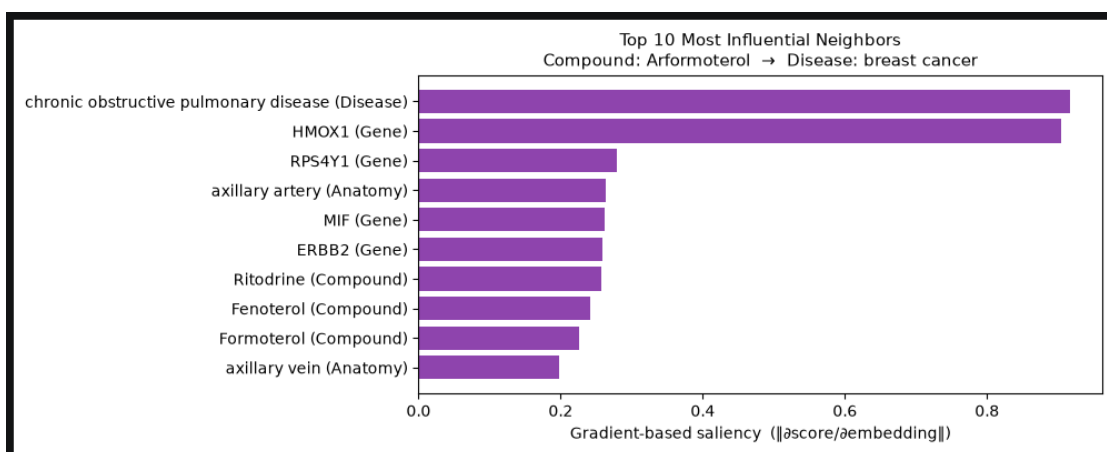


Figure 4.26 Top 10 Most Influential Neighbors — Arformoterol → Breast Cancer

For Desloratadine → asthma, allergic rhinitis (Disease, ≈ 1.62) is the dominant neighbor, more than double the saliency of the next-ranked entity — the largest single-neighbor dominance seen in any of the five pairs — followed by the genes FAM205A and CST1 and the compounds Loratadine, Azatadine, and Amoxapine, all pharmacologically related antihistamine or antihistamine-adjacent compounds.

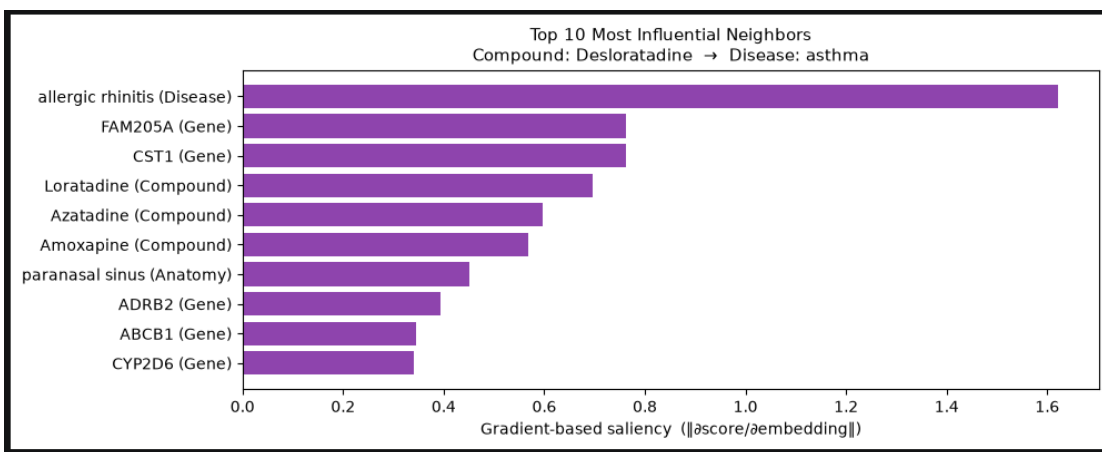


Figure 4.27 Top 10 Most Influential Neighbors — Desloratadine → Asthma

For Cisplatin → Crohn's disease, the influential-neighbor profile looks qualitatively different from the other four pairs: the top ten neighbors are almost entirely gastrointestinal anatomy terms (feces, intestine secretion, superior mesenteric artery, perineum, scrotum, penis, sacro-iliac joint, myenteric nerve plexus) plus two related diseases — ileum cancer, the single top neighbor at ≈ 0.36 , and ulcerative colitis — with no compound or gene appearing in the top ten at all. This indicates the score for this specific pair is driven overwhelmingly by Crohn's disease's own local neighborhood in the graph rather than by any of Cisplatin's neighbors, a genuinely different explanatory pattern from the other four pairs, where the drug's own neighbors (other compounds, genes) always appear alongside the disease's.

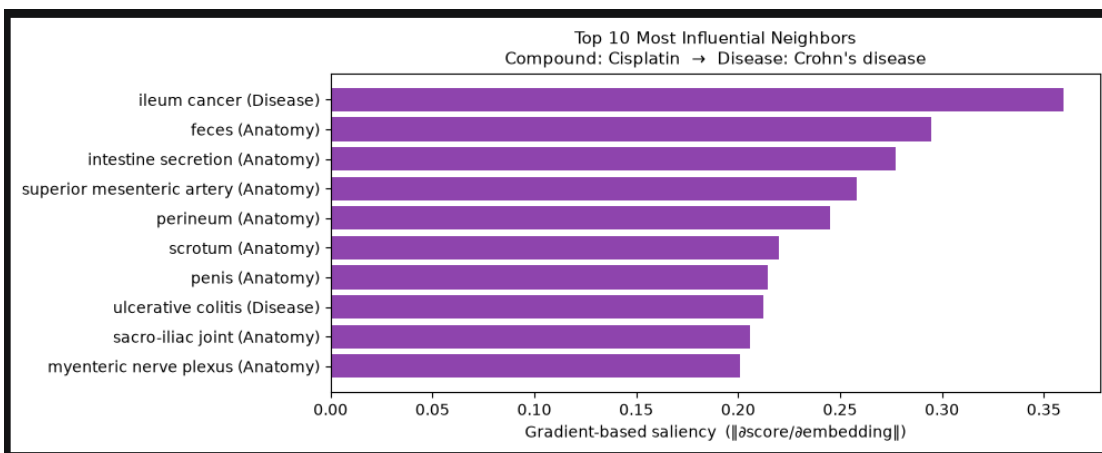


Figure 4.28 Top 10 Most Influential Neighbors — Cisplatin → Crohn's Disease

For Arformoterol → psoriasis, HMOX1 (Gene, ≈ 0.95) and chronic obstructive pulmonary disease (Disease, ≈ 0.77) are again the two most influential neighbors — the same top two entities as Arformoterol → breast cancer above, in the same order — and the same three beta-agonist compounds (Ritodrine, Formoterol, Fenoterol) again appear in the top ten. Because Arformoterol's real 1-hop graph neighbors do not depend on which disease is being scored, this recurrence is expected: it is the disease-independent half of the explanation, and the meaningful comparison is instead which neighbors differ between the

two Arformoterol pairs — here, the disease-side entities (breast cancer's RPS4Y1, ERBB2, and axillary anatomy vs. psoriasis's carpal region, sternoclavicular joint, and elbow).

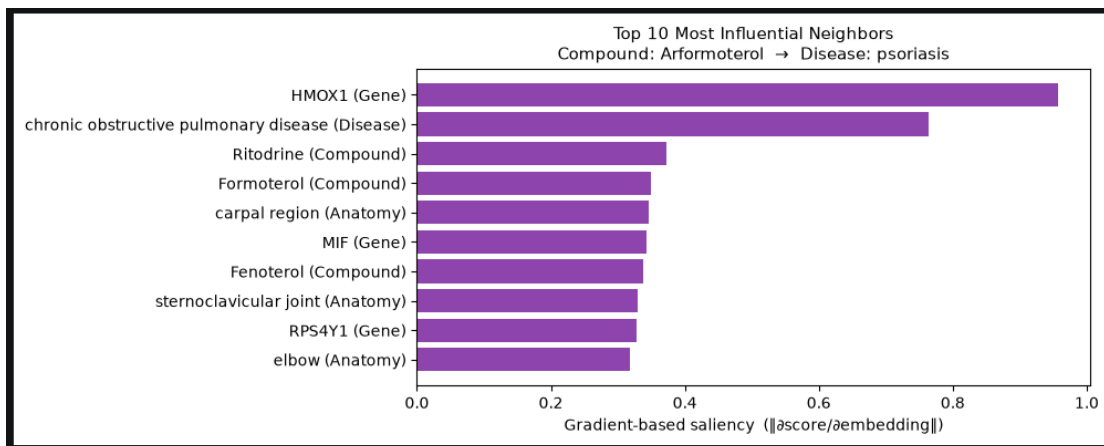


Figure 4.29 Top 10 Most Influential Neighbors — Arformoterol → Psoriasis

4.5.4 Method 3: Relation-Type Importance (Ablation)

Figures 4.30–4.34 take a coarser but complementary view: each of the graph's relation (edge) types is emptied one at a time, the embeddings are recomputed from the resulting graph, and the change in each pair's predicted score is recorded (baseline score minus ablated score; a positive value means that relation type supports the prediction). Because the ablated embeddings depend only on the graph structure and not on which pair is scored, they were computed once and reused for all five pairs, and every removed relation's edges were restored immediately afterward, leaving the evaluation graph unchanged.

For Desloratadine → multiple sclerosis (baseline score = 7.3443), the top-ranked relation is uniquely Compound palliates Disease rather than Compound treats Disease — the only one of the five pairs whose single most important relation is not Compound treats Disease. Disease presents Symptom and Pharmacologic Class includes Compound follow closely behind, with Compound treats Disease itself ranking fourth. Disease localizes to Anatomy also appears twice with opposite signs depending on traversal direction (forward negative, reversed positive), showing that the two directions of the same relation code can carry different, even opposing, information.

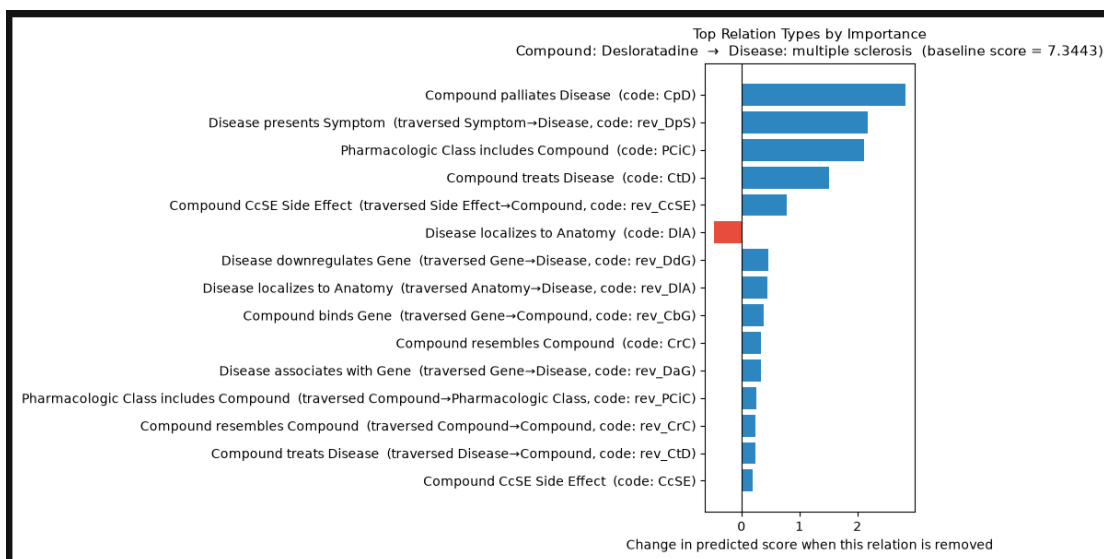


Figure 4.30 Top Relation Types by Importance — Desloratadine → Multiple Sclerosis

For Arformoterol → breast cancer (baseline score = 8.6437), Compound treats Disease is clearly the top relation, followed by Pharmacologic Class includes Compound and Compound palliates Disease — the same top-three ordering as most other pairs — but this pair shows one clear divergence: the traversed Side Effect→Compound relation is negative here, meaning removing it slightly increases the predicted score, the opposite of its effect for Arformoterol → psoriasis below, where the identical relation type is one of the largest positive contributors. Since it is the same drug and the same relation type in both cases, this sign flip is attributable entirely to the disease side of the pair.

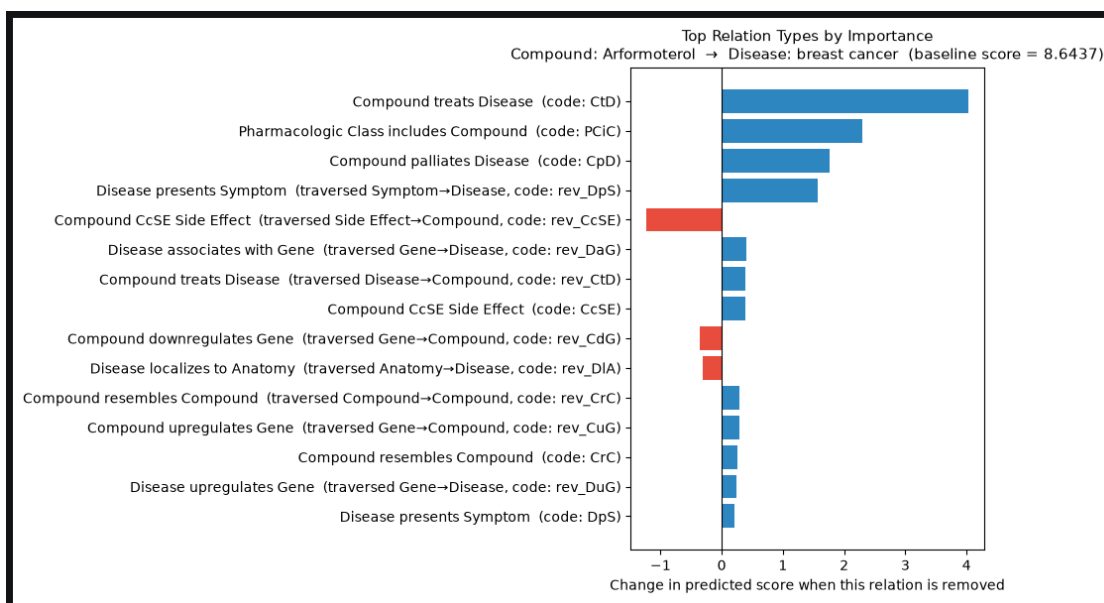


Figure 4.31 Top Relation Types by Importance — Arformoterol → Breast Cancer

For Desloratadine → asthma (baseline score = 16.8042), Compound treats Disease dominates by a wide margin, consistent with this pair's much higher overall score, followed by Compound palliates Disease and Pharmacologic Class includes Compound. This is the

only pair among the five where Disease localizes to Anatomy is clearly negative, again showing that the same relation type's effect is pair-specific rather than fixed.

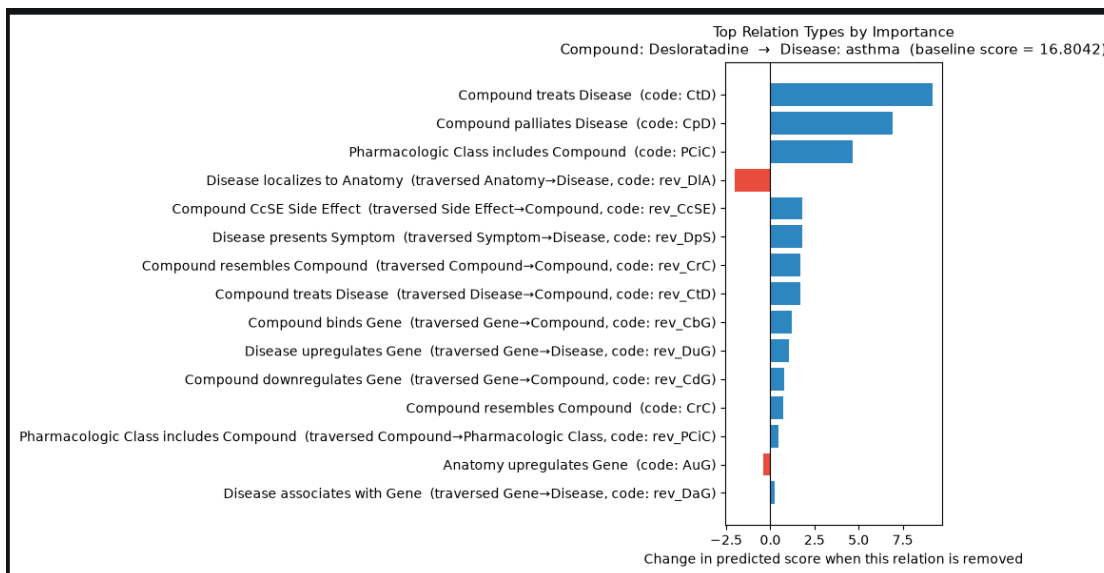


Figure 4.32 Top Relation Types by Importance — Desloratadine → Asthma

For Cisplatin → Crohn's disease (baseline score = 7.9641) is the clearest outlier of the five: Pharmacologic Class includes Compound is the top relation, not Compound treats Disease, which instead ranks near the bottom of the fifteen relations shown and is slightly negative — removing it marginally increases the score. The next three relations by importance are all disease-side (Disease localizes to Anatomy, Disease presents Symptom, Compound palliates Disease), consistent with the gradient-saliency finding above that this pair's explanation is driven mainly by Crohn's disease's own graph neighborhood rather than by Cisplatin's.

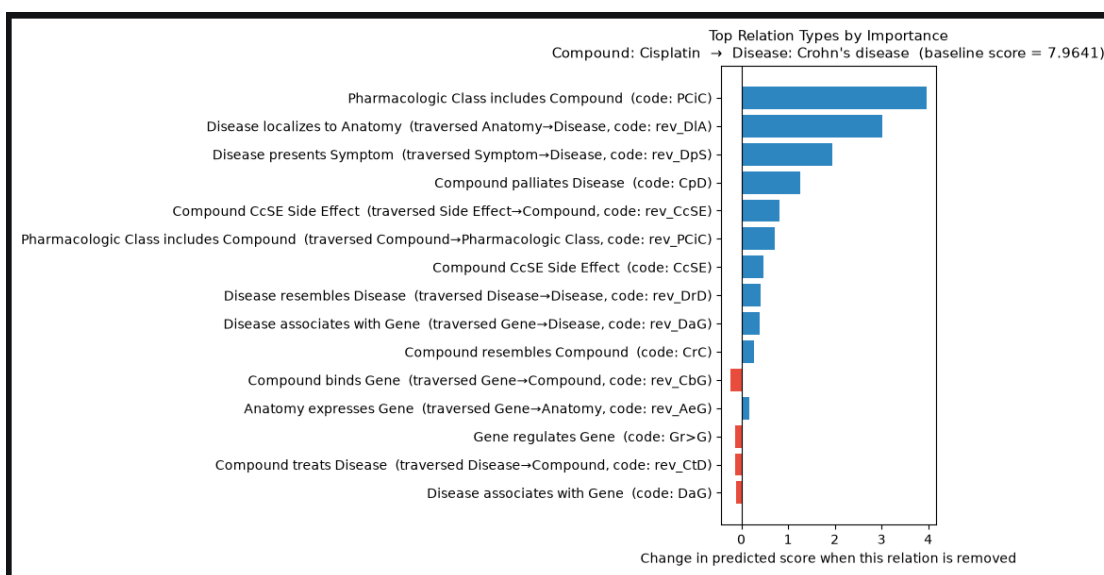


Figure 4.33 Top Relation Types by Importance — Cisplatin → Crohn's Disease

For Arformoterol → psoriasis (baseline score = 7.8781) returns to the more typical pattern seen in three of the other four pairs: Compound treats Disease is the top relation, followed by Pharmacologic Class includes Compound and the traversed Side Effect→Compound relation — the latter positive here, in contrast to its negative sign for Arformoterol → breast cancer above.

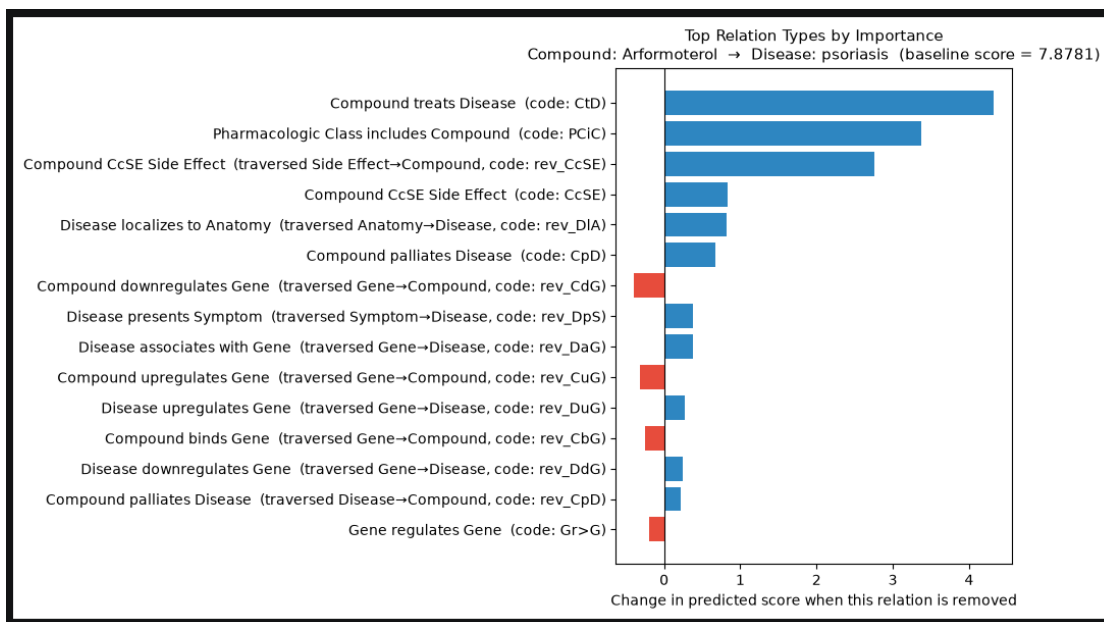


Figure 4.34 Top Relation Types by Importance — Arformoterol → Psoriasis

4.5.5 Method 4: Global / Aggregated Feature Importance Across Multiple Predictions

Figure 4.35 combines Method 1's per-pair dimension attributions into a single heatmap across all five pairs, ranking the twenty dimensions with the largest mean |contribution|. Dimension 115 (multiple sclerosis's top dimension), dimension 53 (both Arformoterol pairs' top dimension), and dimension 59 (Crohn's disease's top positive term and asthma's top negative term) are all visible among the highest-ranked columns, and the checkerboard pattern of red and blue across rows confirms that no single dimension acts uniformly across all five predictions — the same dimension can be strongly positive for one pair and strongly negative for another, exactly as dimension 59 was for Crohn's disease versus asthma in Section 4.5.2.

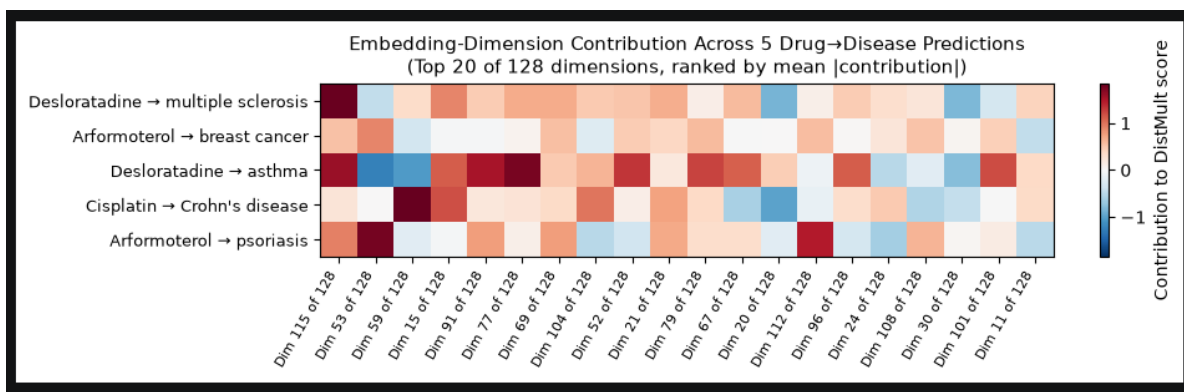


Figure 4.35 Embedding-Dimension Contribution Across 5 Drug→Disease Predictions (Top 20 of 128 Dimensions)

Figure 4.36 performs the equivalent aggregation for Method 3, averaging each relation type's importance (with its standard deviation) across all five pairs. Compound treats Disease is the globally most important relation type on average, but also carries the widest error bar of any relation shown — consistent with Section 4.5.4's finding that its individual per-pair importance ranges from clearly dominant (Desloratadine → asthma) to slightly negative (Cisplatin → Crohn's disease). Pharmacologic Class includes Compound and Compound palliates Disease follow, with visibly narrower error bars, indicating more consistent importance across the five pairs than Compound treats Disease itself.

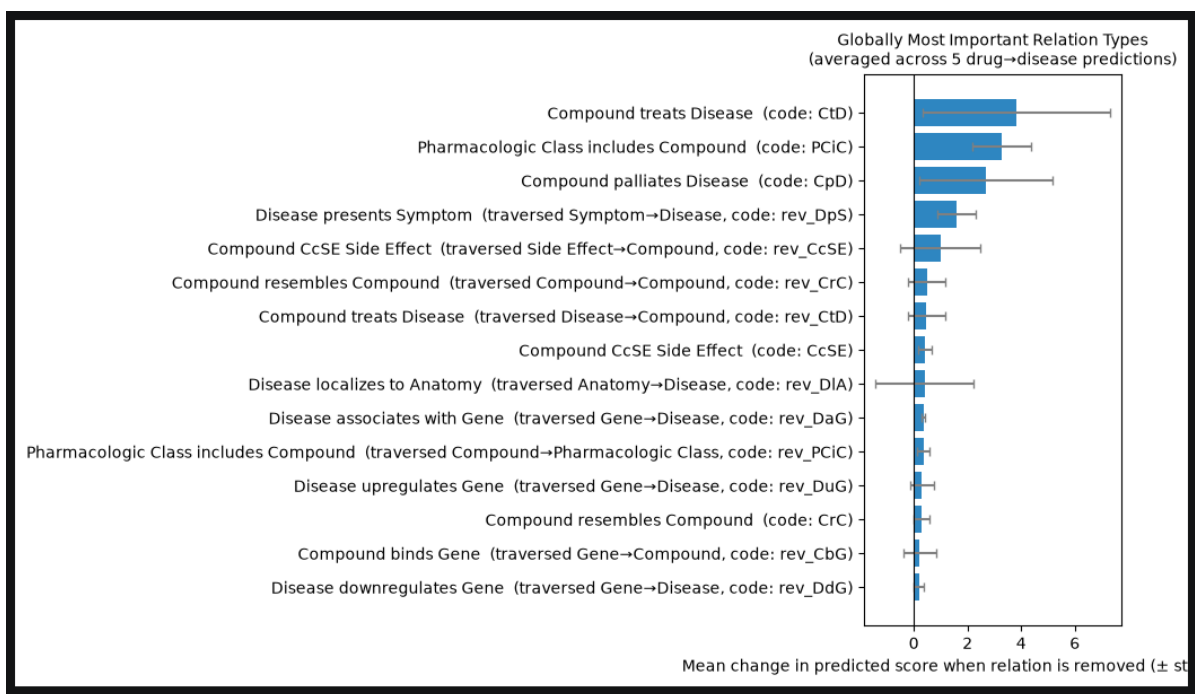


Figure 4.36 Globally Most Important Relation Types (Averaged Across 5 Drug→Disease Predictions)

4.5.6 Summary — Cross-Method Agreement and What This XAI Section Does and Does Not Establish

Table 4.6 collects, for each of the five pairs, the single top-ranked result from each of the three per-pair methods above, as a quick cross-method reference before the discussion that follows.

Table 4.6 Per-Pair XAI Method Agreement Summary — TxGNN

Pair	Score	Top dim (M1)	Top neighbor (M2)	Top relation (M3)
Desloratadine → Multiple sclerosis	7.3443	Dim 115 of 128	allergic rhinitis (Disease)	Compound palliates Disease
Arformoterol → Breast cancer	8.6437	Dim 53 of 128	chronic obstructive pulmonary disease (Disease)	Compound treats Disease
Desloratadine → Asthma	16.8042	Dim 77 of 128	allergic rhinitis (Disease)	Compound treats Disease
Cisplatin → Crohn's disease	7.9641	Dim 59 of 128	ileum cancer (Disease)	Pharmacologic Class includes Compound
Arformoterol → Psoriasis	7.8781	Dim 53 of 128	HMOX1 (Gene)	Compound treats Disease

- The exact dimension-level decomposition (Method 1, Figures 4.20–4.24) and the aggregated heatmap (Figure 4.35) show precisely which of the 128 learned coordinates drive each prediction, with no approximation error, since the DistMult score is by construction their sum; unlike the TreeSHAP dimensions in Section 4.4, however, each TxGNN dimension is a joint drug×disease×relation-weight term rather than a feature belonging to one side alone, so a dimension's effect on the score cannot be attributed to the drug or the disease in isolation.
- Gradient-based neighbor saliency (Method 2, Figures 4.25–4.29) is the layer that most directly parallels the metapath evidence used for XGBoost in Section 4.4.2.4: it names real, trained-on Hetionet entities rather than latent coordinates, and for four of the five pairs it surfaces plausible pharmacological neighbors — related antihistamines for Desloratadine's pairs, related beta-agonists for Arformoterol's pairs. For Cisplatin → Crohn's disease it instead surfaces almost exclusively the disease's own gastrointestinal neighborhood, which, combined with the relation-ablation finding below, suggests this particular prediction leans more on disease-side graph structure than on any specific property of Cisplatin.
- Relation-type ablation (Method 3, Figures 4.30–4.34, aggregated in Figure 4.36) shows that Compound treats Disease is the most important relation type on average but the least consistent one: it is the top-ranked relation for three of the five pairs, a secondary relation for one, and mildly negative for Cisplatin → Crohn's disease, whose top relation is instead Pharmacologic Class includes Compound. Reporting

this disagreement directly, rather than only the average, follows the same principle applied to the Run 1 vs Run 2 comparison in Table 4.4 for XGBoost.

- Across all four methods, the two drugs that appear twice in this section — Desloratadine and Arformoterol — give a direct way to separate drug-specific signal from disease-specific signal: their shared top embedding dimension, shared top graph neighbors, and (for three of the four applicable pairs) shared top relation type all recur across both of their predictions, while the specific secondary neighbors and the sign of at least one relation type (the traversed Side Effect→Compound relation for Arformoterol) differ by disease. This is consistent with the model combining a genuine drug-level representation with a disease-specific modulation, rather than simply memorizing five independent pairs.

As with the XGBoost explanations in Section 4.4.3, none of the four methods above substitutes for experimental validation: they establish which latent dimensions, which real graph neighbors, and which relation types the trained TxGNN model itself relied on for each of these five predictions, and where that reliance is consistent or inconsistent across pairs — not whether the underlying pharmacology is correct. Cisplatin → Crohn's disease in particular, whose explanation diverges from the other four pairs across all three per-pair explanation methods, is a candidate this section flags as resting more on graph structure than on a specific compound-level signal, and should be weighted accordingly relative to the other four candidates until checked against external evidence.

4.6 Deployment

The system is deployed as a three-tier web application that keeps the user interface, request orchestration, and model inference in independent services, so each layer can be restarted, scaled, or replaced on its own.

Frontend (React): A React single-page application lets a user search for a drug or a disease, choose a model, and view the ranked predictions, supporting evidence, and an interactive graph of the underlying relationships.

Backend Gateway (Node.js / Express): A lightweight Express server is the single entry point the frontend talks to. It validates incoming requests, forwards them to the prediction service, and reshapes the response into the format the UI expects, while also handling CORS and centralising error handling.

Prediction Service (FastAPI): A Python FastAPI service wraps the trained XGBoost/DREAMwalk, TxGNN, and ensemble models together with the literature-evidence pipeline. It exposes internal endpoints for predictions, drug/disease lookup, and PubMed evidence, and is only ever called by the Express gateway, never directly by the browser.

In development the two backend services run locally on separate ports (Express on 8000, FastAPI on 8001) and are started together with a single launcher script; the same architecture can be containerised and placed behind a reverse proxy for production without changing the frontend code.

Chapter (5)

Conclusion and Future Work

This project set out to show that combining heterogeneous knowledge-graph learning with explainable AI can turn Hetionet's biomedical relationships into ranked checkable drug-repurposing hypotheses rather than opaque scores. Two complementary models were built and evaluated: a DREAMwalk-based pipeline that embeds the graph with a semantic, teleport-guided random walk before scoring drug–disease pairs with XGBoost, and a TxGNN heterogeneous graph neural network trained end-to-end under a leakage-free evaluation protocol. Both pipelines were wrapped in a shared explainability layer — SHAP-based feature attribution together with an independent metapath-evidence check — so that every top-ranked candidate is accompanied by a stated, verifiable reason rather than a bare probability.

To make these results usable beyond a notebook, the models were packaged behind a FastAPI prediction service and exposed through a React and Node.js web application, letting a user query a drug or disease and review the ranked candidates, supporting genes and pathways, and literature evidence behind each prediction in one place.

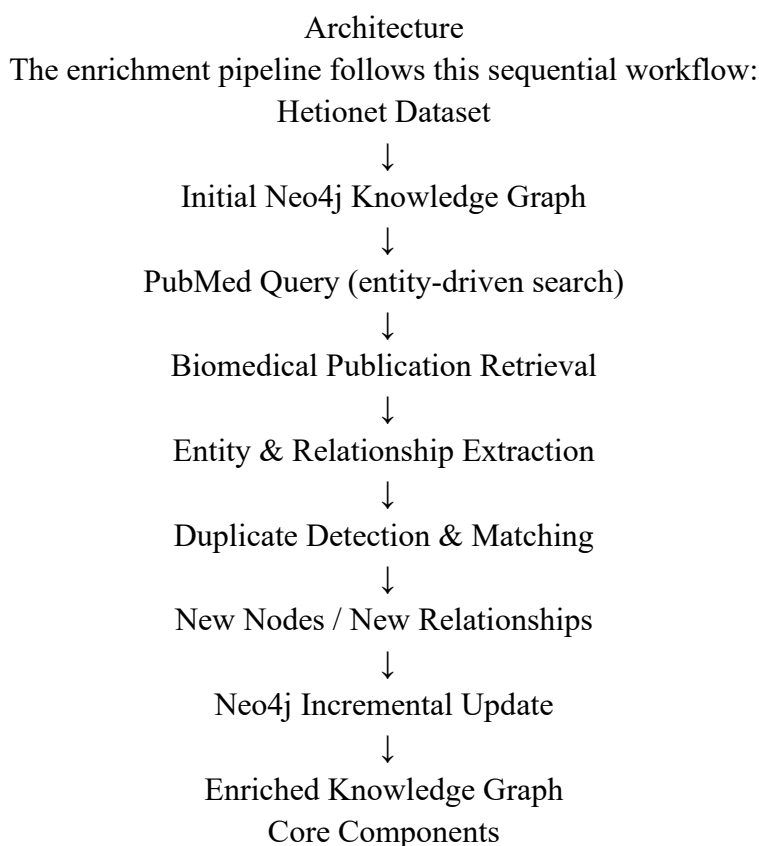
As emphasised throughout Chapter 4, these results are hypotheses to test, not clinical conclusions: the models were evaluated against Hetionet alone, no wet-lab or clinical validation was performed, and literature evidence was retrieved from PubMed metadata rather than curated by domain experts. With that scope in mind, the main directions for future work are:

- **Expand Data:** Integrate additional biomedical databases beyond Hetionet to increase graph coverage and capture more drug–disease relationships.
- **Advanced GNNs:** Explore more advanced heterogeneous GNN architectures and attention mechanisms to improve representation learning and prediction performance.
- **DYNAMIC KG & RAG-BASED EVIDENCE RETRIEVAL:** Continuously update the knowledge graph and extract real-time biomedical evidence from PubMed using advanced RAG pipelines
- **Experimental Validation:** Prioritise the highest-ranked novel drug–disease candidates for literature-based, preclinical, and eventually clinical validation.

Current work

PubMed-Based	Knowledge	Graph	Enrichment
Status:	Under	Active	Development
This component is currently under active development and represents a planned extension to the core drug repurposing framework. The foundational architecture has been designed and implementation is ongoing. Full integration with the prediction pipeline is intended for a subsequent release.			

Overview To further strengthen the biological evidence available to both the DREAMwalk and TxGNN models, this component introduces a continuous knowledge graph enrichment pipeline that augments the static Hetionet graph with dynamically retrieved biomedical literature from PubMed. While Hetionet v1.0 provides a comprehensive and structured initial knowledge base, it represents a fixed snapshot of biomedical knowledge as of its release. Biomedical research advances rapidly — new drug-gene associations, disease mechanisms, and therapeutic hypotheses are published continuously. This component addresses that limitation by treating the knowledge graph as a living structure that grows incrementally as new evidence emerges.



Neo4j as the Graph Store: The Hetionet knowledge graph is loaded into Neo4j, a native graph database that represents biological entities as nodes and their associations as typed relationships. Neo4j is selected specifically because biomedical knowledge is inherently

relational — a drug targeting a gene that is associated with a disease is a multi-hop path that graph databases traverse naturally and efficiently. PubMed Retrieval: Queries are constructed from the existing entity set (compound names, disease names, gene symbols) and submitted to the PubMed API. Retrieved publications are processed to extract candidate entities and relationships not currently present in the Hetionet graph. Entity and Relationship Matching: Before any insertion, extracted entities are matched against existing Neo4j nodes using normalized identifiers. This prevents duplicate node creation when the same biological entity appears across multiple publications. The same deduplication logic applies to relationships — a Drug → treats → Disease edge already present in Hetionet is never duplicated regardless of how many PubMed sources confirm it.

Incremental Update: Only genuinely new nodes and relationships are written to Neo4j. This incremental strategy preserves the integrity of the original Hetionet graph while continuously extending it with newly discovered biological evidence. Contribution to the Pipeline Once completed, this component will provide both DREAMwalk and TxGNN with a richer knowledge foundation — one that reflects current biomedical literature rather than being limited to Hetionet's original release. Drug-gene and gene-disease connections discovered in recent publications will become available as new graph edges, directly expanding the evidence base from which random walks and GNN message passing draw their signal.

Current Development Status

Sub-component Status

Neo4j setup and Hetionet loading	Complete
PubMed API query pipeline	In Progress
Entity extraction and NER	In Progress
Duplicate detection logic	In Progress
Neo4j incremental update	Pending
Integration with prediction pipeline	Pending

ملخص المشروع

يُعد إعادة توظيف الأدوية (Drug Repurposing) بديلاً استراتيجياً مهماً لاكتشاف الأدوية الجديدة من الصفر (de novo drug discovery) ، وذلك من خلال تحديد استخدامات علاجية جديدة للمركبات الدوائية المعتمدة والموجودة بالفعل، مما يساهم في تقليل مدة التطوير والتكلفة ومخاطر فشل الدواء خلال المراحل السريرية. ومع ذلك، لا تزال عملية إعادة توظيف الأدوية بشكل منهجي تمثل تحدياً، نظراً إلى أن العلاقات المعروفة بين الأدوية والأمراض شديدة الندرة، حيث تمثل فقط 0.36% من إجمالي مساحة البحث للمركبات والأمراض في Hetionet v1.0، بالإضافة إلى ارتفاع درجة عدم تجانس البيانات البيولوجية، وافتقار العديد من النماذج التنبؤية إلى قابلية التفسير.

يقدم هذا المشروع إطاراً متكاملًا قائمًا على الذكاء الاصطناعي (AI) لإعادة توظيف الأدوية، معتمدًا على Hetionet، وهي Heterogeneous Biomedical Knowledge Graph تدمج بيانات من 29 قاعدة بيانات، وتشمل 11 نوعاً من العقد (Node Types) ، و 24 نوعاً من العلاقات (Relation Types) ، و 47,031 عقدة وأكثر من 2.25 مليون علاقة. ولتقليل الاعتماد على نموذج واحد، يجمع الإطار بين مكوّنين تنبؤيين مستقلين من الناحية المعمارية.

يعتمد المكوّن الأول على DREAMwalk مع XGBoost ، حيث يتم إنشاء Node Embeddings بُعدها 128 من خلال Semantic Information-Guided Random Walks بطول 100، مع تنفيذ 20 جولة مشي لكل عقدة، واحتمالية انتقال (Teleportation Probability) قدرها (p=0.5). ويتم اشتقاق احتمالات الانتقال من خلال Vectorized Jaccard Similarity Matrices محسوبة لكل نوع من أنواع الكيانات. بعد ذلك، يتم ترميز تسلسلات المشي باستخدام Heterogeneous Skip-gram Word2Vec، بينما يتم تمثيل أزواج الدواء والمرض باستخدام Concatenated Embeddings بُعدها 256، ثم تصنيفها باستخدام XGBoost بعدد 300 شجرة (Estimators) ، وأقصى عمق 6، ومعدل تعلم 0.001. ويتم تقييم هذا المكوّن باستخدام بروتوكول 10-Fold Stratified Cross-Validation خالٍ من تسرب البيانات (Leak-Free).

أما المكوّن الثاني، TxGNN، فهو Heterogeneous Graph Neural Network يتم تدريبه من البداية إلى النهاية (End-to-End) باستخدام PyTorch Geometric. ويستخدم Type-Specific Learnable Node Embeddings بُعدها 128، بالإضافة إلى مُشغّر مكوّن

من طريقتين من HeteroConv باستخدام عمليات GraphSAGE الخاصة بكل علاقة، و DistMult Link Scoring لعلاقات CtD و CpD ويعتمد التدريب على Auxiliary و Dynamic Negative Sampling، و BCEWithLogitsLoss Contrastive Metric Loss، مع استخدام 5-Fold Stratified Cross-Validation خالي من تسرب البيانات.

بعد ذلك، يتم دمج النموذجين من خلال Weighted Ensemble، بحيث يتم تخصيص 55% من الوزن لـ DREAMwalk/XGBoost و 45% لـ TxGNN. وتوفر آليات التعلم التكاملية للنموذجين، والتي تعتمد على Walk-Based Co-Occurrence من ناحية و Iterative Graph Message Passing من ناحية أخرى، تمثيلات متنوعة تساعد في تقليل التنبؤات الإيجابية الكاذبة. (False Positive Predictions).

تم تحقيق قابلية التفسير (Interpretability) من خلال SHAP-Based Feature Attribution على مصنف XGBoost، مع عرض النتائج في صورة Interactive Waterfall Charts ضمن واجهة ويب مخصصة لدعم اتخاذ القرار، وتدعم الاستعلامات من Drug-to-Disease و Disease-to-Drug ويتم ترتيب التنبؤات وفقًا لـ Ensemble Confidence، مع إثرائها بمجموعات الجينات الداعمة (Supporting Gene Sets) والأدلة المباشرة من Hetionet ومعلومات المسارات البيولوجية (Biological Pathway Information)، بالإضافة إلى XAI Assistant يقوم بتحويل مخرجات SHAP إلى تفسيرات مفهومة بلغة بشرية.

حقق خط أنابيب DREAMwalk/XGBoost قيمة AUROC 0.938 و Accuracy 0.873 على معيار Hetionet، بزيادة قدرها 16.8% مقارنةً بالنماذج المرجعية المتقدمة-State-of-the-Art Baselines). كما أظهرت دراسات الحالة الواقعية التي استهدفت مرض ألزهايمر (Alzheimer's Disease) وسرطان الثدي (Breast Carcinoma) قابلية عملية لتطبيق الإطار المقترح.

وبشكل عام، يوضح هذا الإطار كيف يمكن للجمع بين Heterogeneous Knowledge و Graphs و Semantic Random Walks و Graph Neural Networks و Ensemble Learning و Explainable AI أن يدعم عملية أكثر دقة ومنهجية وقابلية للتفسير لإعادة توظيف الأدوية.



الكلية الفنية العسكرية



إعادة توجيه استخدام الأدوية باستخدام الذكاء الاصطناعي

رسالة مقدمه من

جيهان عبد الحميد

منار محمد

بسنت عادل

رحمة جمال

رويدا صلاح

باسنت محمود

رسالة مقدمة استكمالاً لمتطلبات الحصول على

دبلومة Digilians 9 Months

مسار الذكاء الاصطناعي التطبيقي و تحليل البيانات

يعتمد من لجنة الممتحنين

التوقيع

الإسم



الكلية الفنية العسكرية



إعادة توجيه استخدام الأدوية باستخدام الذكاء الاصطناعي

رسالة مقدمه من
جيهان عبد الحميد
منار محمد
بسنت عادل
رحمة جمال
رويدا صلاح
باسنت محمود

رسالة مقدمة استكمالاً لمتطلبات الحصول على دبلومة Digilians لمدة تسعة أشهر في الذكاء الاصطناعي
التطبيقي وتحليل البيانات

تحت إشراف
اللواء.أ.د.جوده إسماعيل

أغسطس 2026